

Gender agreement mismatch in sandwiched coordination

Marija Brašić, University of Nova Gorica, marija.brasic@gmail.com

This paper reports on an experimental study on noun-verb agreement in so-called *sandwiched configurations*, where a coordinated subject (Goal) is situated between two participles (Probes). As reported by Marušič & Nevins (2020), in such configurations the two conjuncts of a coordinate subject control agreement with different probes, each one with the linearly closest probe – the first conjunct with the probe preceding the subject and the second conjunct with the probe following the subject. This study in addition confirms that two participles show agreement with the single, hierarchically highest conjunct, as well as having the first probe reflecting gender features of the closest (hierarchically highest) noun and the second one displaying default gender agreement. Hence, the configurations mirror exactly the agreement strategies found in simple, pre- and post- verbal, coordinate structures. The acceptability rating experiment of Marušič & Nevins (2020) is conceptually replicated in Bosnian-Croatian-Montenegrin-Serbian (BCMS), with results confirming their study and their conclusions. The distributed model of agreement (Arregi & Nevins 2012) considers a two-step Agree operation, the second step of which, Agree-Copy, taking place post-syntactically, before or after linearisation, thus allowing Linear order to have a direct effect on agreement.

Keywords: coordination, gender, conjunction, noun-verb agreement

1 Agreement under coordination with gender-mismatched conjuncts

Agreement patterns with coordinate structures are amongst the most intricate agreement phenomena. South Slavic languages offer a productive research domain for conjunct agreement due to their rich verbal morphology, which expresses agreement with either one or both conjoined nouns in person, number, and gender, phi (ϕ)-feature agreement (cf. Corbett 1983a; Marušič et al. 2015; Willer-Gold et al. 2016; 2018 a. o.).

Agree is traditionally considered one of three basic syntactic operations, along with Merge and Move (Chomsky 2000; 2001). Merge constructs new syntactic elements, move rearranges elements within a complex syntactic element while Agree dictates the two operations by checking and valuing various features. In minimalist syntax, Agree is understood as an operation between a goal and a probe. The ϕ -features are probed for by the verbal agreeing head. Agreement is established when the nouns (conjuncts of the &P) assign a value to the Participle based on their ϕ -features.

Coordinated subjects already present a problem in case the two individual conjuncts don't share their gender features. Gender features are not "additive" like number features so there is no principled way to determine what is the common gender of a coordination of a neuter and a feminine noun. As argued for by Marušič et al. (2007; 2015), Willer-Gold et al. (2016; 2018), to name a few, in such cases the verbal probe has three options, it can agree with the linearly closest conjunct, it can agree with the structurally higher conjunct inside the &P, or it can agree in masculine gender which is arguably the default gender value for plural nouns in South Slavic (1).¹

¹ Unless specified otherwise, all BCMS judgments are those of the author.

(1) Bosnian-Croatian-Montenegrin-Serbian

[*Nagrada i priznanja*] *su* *pripremljene/a/i*.
 award-F.PL and acknowledgments-N.PL AUX-PL prepared-F/N/M.PL
 ‘Awards and acknowledgments are prepared.’

A further problem is when there are two verbal probes in a sentence such that they appear on both sides of the &P in a sandwiched configuration. Notice that in such set-up the closest conjunct is different for the two verbal probes, as in (2). The central question of this paper is what happens if different goals with distinct ϕ -features are most accessible to the two verbal probes of the same clause, and how to account for such agreement patterns.

(2) BCMS

Za petak su bile [nagrada i priznanja] pripremljena.
 for friday AUX-PL were-F.PL awards-F.PL and acknowledgments-N.PL prepared-N.PL
 ‘Awards and acknowledgments had been prepared for Friday.’

In the following subsection, we elaborate on coordinate structures producing various agreement patterns. Previous accounts of conjunct agreement are reviewed in Section 2. Using the most recent distributed model in this study to account for sandwiched agreement patterns in Bosnian-Croatian-Montenegrin-Serbian (BCMS), Section 3 summarises recent advances in modelling agreement. Sections 4 and 5 define predictions and present an acceptability judgment rating experiment on BCMS native speakers, respectively. Section 6 outlines the discussion of results and implications, and we conclude in Section 7.

1.1 The complexity of coordinate structures

A hierarchical structure for coordination phrase (as given in Figure 1) is adopted in this study, where the coordination itself is taken to be a Boolean Phrase (&P).

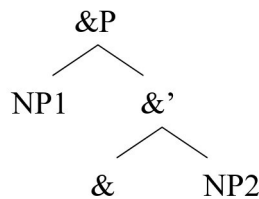


Figure 1: Hierarchical structure adopted for conjunction phrase.

In BCMS, auxiliaries agree in person and number, whereas finite verbs can agree in person, number, and gender. Participles, used in complex active and passive past and future formulations, behave like adjectives (agreement in number, gender, and case), and agree in number and gender, with gender distinction in plural. Table 1 displays the participle endings in BCMS.

Number	Masculine	Feminine	Neuter
Singular	Ø	-a	-o
Plural	-i	-e	-a

Table 1: Participle agreement endings in BCMS.

In the absence of any gender conflict between two conjuncts, &° can calculate a common gender value for the entire coordination (cf. Marušič et al. 2015). But when nouns with different gender and number feature values are conjoined some sort of resolution is needed (Corbett 1983b). This results in an underspecified gender value on &P. We are interested in the processes that help resolve this conflict. As mentioned above, different subject-verb agreement patterns are observed with coordinate constructions in BCMS: closest conjunct agreement, CCA (3a), hierarchically highest conjunct agreement, HCA (3b), and default masculine agreement, def (3c).

CCA (LCA)

- (3) a. [*Nagrade i priznanja*] *su* *pripremljena*.
award-F.PL and acknowledgments-N.PL AUX-PL prepared-N.PL
‘Awards and acknowledgments are prepared.’

HCA

- b. [*Nagrade i priznanja*] *su* *pripremljene*.
award-F.PL and acknowledgments-N.PL AUX-PL prepared-F.PL
‘Awards and acknowledgments are prepared.’

def

- c. [*Nagrade i priznanja*] *su* *pripremljeni*.
award-F.PL and acknowledgments-N.PL AUX-PL prepared-M.PL
‘Awards and acknowledgments are prepared.’

Examples (3a-c) illustrate agreement patterns with pre-verbal (PreV) subjects. When subjects are post-verbal (PostV) the three strategies have a different outcome. CCA and HCA both result in agreement with the first conjunct (which is both highest inside the &P and linearly closest to the verbal probe), (4a), while agreement with the second conjunct results in ungrammaticality (4b). Further, default agreement is degraded, (4c), though there is some disagreement in the literature on the availability of this pattern for BCMS, such as Willer-Gold et al. (2016) vs. Murphy & Puškar (2018).

CCA (HCA)

- (4) a. *Pripremljene* *su* [*nagrade i priznanja*].
prepared-F.PL AUX-PL award-F.PL and acknowledgments-N.PL
‘Awards and acknowledgments are prepared.’

*LCA

- b. **Pripremljena su [nagrade i priznanja].*
prepared-N.PL AUX-PL award-F.PL and acknowledgments-N.PL
'Awards and acknowledgments are prepared.'

??def

- c. ??*Pripremljeni su [nagrade i priznanja].*
prepared-M.PL AUX-PL award-F.PL and acknowledgments-N.PL
'Awards and acknowledgments are prepared.'

As shown, CCA and HCA patterns are possible with both SV and VS orders. The def agreement pattern is judged as grammatical in PreV (SV) but in PostV (VS) as degraded. Following Marušič and Nevins (2020), I use the term sandwiched configurations for coordinate subjects in between two agreeing elements. These elements need not be only verbal elements, but include also a complementizer (C) and a verb (V), or an adjective (Adj) and V.

An increasing number of studies are being conducted to account for agreement in pre-verbal, post-verbal, and subsequently sandwiched configurations in various languages. The effect of a coordinate subject on agreement between C and V in Dutch is discussed by van Koppen (2005; 2007) and in Polish by Citko (2004; 2018). Bhatt & Walkow (2013) discuss conjunction agreement in Hindi/Urdu under the head-complement relation. Empirical and experimental studies in Slovenian consider conjunction phrases that are placed before and after the predicate (Marušič et al. 2007; Marušič & Nevins 2010; Marušič et al. 2015), in between two Vs (Marušič & Nevins 2020), and cases of exclusive disjunction (Marušič & Shen 2021). The work on BCMS comprises cases where Vs precede and follow a coordinate subject, in both conjunction and disjunction, as well as when Adj and V agree with a coordinate phrase (Aljović & Begović 2016; Arsenijević & Mitić 2016a,b; Arsenijević et al. 2019; Begović & Aljović 2015; Čordalija et al. 2016, 2020; Franks & Willer-Gold 2015; Leko 2015a,b; Murphy & Puškar 2018; Mitić & Arsenijević 2019a,b; Willer-Gold et al. 2016; 2018 a. o.). Begović & Aljović (2015) and Aljović & Begović (2016) put forward the use of a sandwiched setup in the Pluperfect tense in BCMS to address conjunct agreement. We will leave other combinations for future research and concentrate on double agreement patterns with verbal participles in this study. The subsequent section examines previous efforts to explain conjunct agreement.

2 Earlier attempts to account for conjunct agreement

This section discusses previous approaches deriving the existing conjunct agreement patterns, followed by the account adopted in this study. Single conjunct agreement, also known as partial conjunct agreement or conjunct sensitive agreement, presents problems to traditional syntactic approaches to agreement phenomena. Conjunct agreement strategies in Western South Slavic languages include closest conjunct agreement (CCA), hierarchically highest (HCA) or first (FCA)

conjunct agreement, and resolved or default (def)², as previously established based on work analysing data from Slovenian and BCMS.

2.1 Syntactic accounts (Bošković 2009; Murphy & Puškar 2018)

Bošković (2009) argues that number and gender probes are a single probe from a Part head with unvalued features (but see Bejar 2003; Marušič et al. 2007 for the opposite view). The Part head probes for both, number and gender, features. It does not stop when it matches &P for the number, since it is not specified for gender. When Part matches NP₁, the probing stops due to (hierarchically) Closest Agree, having both features of Part found a match. Therefore, we are talking about multiple attempts with gender percolation, according to Bošković's approach, developed from Hiraiwa's (2005) Multiple Agree with &P and NP₁ (Bošković 2009: p. 19)³.

The current feature-checking system of Bošković is valuation-driven, following strict c-command. The crucial assumption for the analysis of Bošković (2009) is that the gender feature of BCMS nominals is valued and uninterpretable and that such features undergo deletion as soon as they undergo Match, i.e. as soon as they are targeted by a probing operation. BCMS nominals have a grammatical gender. They are assigned gender arbitrarily, gender being a grammatical feature without semantic import, according to Bošković. Unvalued uninterpretable features are to be valued before deletion and valued uninterpretable features can be simply deleted (at LF).

What about interpretable features (such as number features), that receive interpretation at LF (Chomsky 1995)? Even some gender features, such as feminine on the noun *žene* 'women' and neuter on the noun *djeca* 'children', have a clear semantic motivation. These nouns in BCMS have a true gender. Bošković treats them as exceptions (see Bošković 2009: §3.3, p. 17). Possibly some semantic motivation includes exclusively natural gender and hybrid nouns, though that makes things too complicated and undesirable. Additionally, hybrid nouns in BCMS are assigned a grammatical gender. However, in the singular form, they require the true gender to be reflected in verbal or adjectival morphology. For more information on gender mismatch and when it is blocked, please refer to the work of Puškar (2018). It is suggested that the natural masculine gender is syntactically absent when the adjective agrees with grammatical gender. Other relevant sources include Despić (2017), Puškar (2017), Murphy, Puškar & Naranjo (2018), Wechsler & Zlatić (2003), and references therein. The gender subsystem in the resolution proposed by Pražmowska (2016) is depicted in Table 2. It demonstrates the treatment of gender specification for nominals, aiming to accurately represent BCMS nouns.

grammatical + interpretable gender		uninterpretable gender
masculine	feminine	human/non-human

Table 2: Pražmowska (2016).

² The difference between the two was discussed in Bošković (2009: Footnote 4), Willer-Gold et al. (2016), and Murphy & Puškar (2018) for BCMS, and Marušič & Shen (2021) for Slovenian, treating them as different strategies.

³ And also partially adopting Bejar (2003). The reader is referred to Bošković (2009) for detailed explanation.

Rather than treating a portion of data as exceptions, it is more desirable to be able to provide an account that can capture as much data as possible. Two important aspects of the Bošković's account are: (i) it is not interpretability but valuation that drives feature checking in syntax, and (ii) the probe is not separate for number and gender. The preverbal HCA pattern defies views of the Bošković's narrow syntactic account. According to his account, preverbal HCA (or distant FCA in his terminology) is predicted to be ungrammatical (Bošković 2009: p. 16, Footnote 23). Yet, such syntactic account was demonstrated to be untenable, as HCA was identified in the subsequent studies of the post-syntactic approach in Slovenian (Marušič et al. 2015) and BCMS (Willer-Gold et al. 2016; 2018). These studies experimentally confirmed pre-verbal HCA.

The sandwiched configurations, termed as double participle constructions (Bošković 2009: Footnote 21), are considered a less favoured choice amongst native speakers of BCMS. Beside questioning such configurations in the first place, Bošković also assumed that the same element can control both gender and number. The derivation under discussion then cannot yield a grammatical output of double HCA (CCA+HCA) and double def (two of the four patterns to be experimentally tested). Furthermore, the gender specification of the individual conjuncts dealing with LCA and FCA in pre- and post-verbal configurations, respectively, is considered authentic CCA, adopting adjacency in the post-syntactic component.

This study will focus specifically on experimentally testing this aspect: (i) whether sandwiched constructions are perceived as acceptable constructions in general, and (ii) if found that separate elements control gender (hierarchically highest conjunct, linearly closet conjunct, and resolution default on &P), while keeping number constant, on participles on both sides, we are one step closer to support the claim that each side probing is independent operation, and subsequently that gender and number are split ϕ -probes (having same number feature on probes but different gender feature values). Elaboration is offered in the subsequent sections of the paper.

Murphy & Puškar (2018) supports the strict syntactic approach positing that a single conjunct agreement is agreement with the conjunction phrase. They proposed that linearly conditioned agreement can be analysed as the result of different derivations inside the conjunction phrase since the order in which syntactic operations (upward and downward Agree (Probe, Match, and Value), alongside Move and Merge) apply is in principle free. The agreement with the conjunct phrase inherited the features of only one of its conjuncts, claim Murphy & Puškar. Important to mention is that according to Murphy & Puškar (2018), the given order of operations remains fixed for later cycles, namely post-syntactic stages. Nonetheless, they do resort to underspecification in answering how are conflicting values resolved. Their approach to resolution is based on constraint-driven impoverishment of conflicting gender values on the Part head in a 'late insertion' approach to morphology such as Distributed Morphology (DM; Halle & Marantz 1993; 1994; 1997; Embick & Noyer 2001; 2012), where masculine is the least marked gender. Crucially, the resolution is then a distinct process to default agreement (Murphy & Puškar 2018: p. 1233), in line with Marušič & Shen (2021). Speakers can always choose the & head with a pre-specified masculine gender value. However, the &P node is not specified for a gender value and thus Part probes inside to find the first conjunct (Murphy & Puškar 2018: p. 1240), directly comparable to the Peeking grammar of Marušič et al. (2015), that we adopt in this study and is discussed in Section 3. Moreover, Murphy & Puškar (2018: p. 1241) do acknowledge the genuine highest conjunct agreement in the post-verbal

configuration, and not &P inheriting the features of the first conjunct, which they show on the BCMS cases of single highest/first conjunct agreement (Murphy & Puškar 2018: p. 1242). Puškar & Murphy (2015) and Murphy & Puškar (2018) report five options as grammatical, with both upward and downward agree available: HCA, LCA, and def pre-verbally, and HCA and def post-verbally.

In their analysis of sandwiched cases, Murphy & Puškar (2018) propose the deactivation property of a conjunct during the derivation of double CCA. Double HCA, experimentally confirmed in this study, directly challenges such an account of deactivation, as features of the single conjunct are morphologically realised on both participles. The purpose of the survey of existing accounts is to see whether they can empirically capture as many possible configurations and agreement patterns as feasible to achieve a more simplified account without making too many additional assumptions.

2.2 Post-syntactic accounts (Marušić et al. 2015; Aljović & Begović 2016⁴)

The post-syntactic approaches focus on the role of linear proximity or linear order in conjunct agreement. It differs from the strict syntactic approach by placing greater emphasis on the later stages of derivation. Such proposals are supported by various studies, including those of Benmamoun, Bhatia & Polinsky (2009), Bhatt & Walkow (2013), Marušić et al. (2007; 2015), to name a few. The accounts propose that conjunct agreement involves the copying of features at the Phonological level (PF) via two-step Agree. The model comprises of two parts: a syntactic part and a post-syntactic part, Agree-Link and Agree-Copy, respectively. Agree-Link involves a probe, such as Infl⁰ or v⁰, that searches for and establishes a connection with a goal, which can be a single noun phrase (NP) or several goals. Agree-Copy is the process by which content from the goal or goals is shared with or copied onto the probe. Agree-Copy at PF can copy previously linked feature values before linearisation, resulting in HCA, or after vocabulary items are inserted, copying the last conjunct feature values (LCA). Middle conjunct(s) agreement with multiple conjuncts is confirmed as untenable (for BCMS, see Arsenijević & Mitić 2016a; Murphy & Puškar 2018), and will not be discussed in the present study.

Marušić et al. (2015) showed that single conjunct agreement (agreement with only one conjunct in coordination) in gender is conditioned by consistency in number, using the combination N_{PL} & F_{SG} (but see Moskovljević 1983 and Bojović 2003 mentioned in Arsenijević & Mitić 2016a for the opposite stand). This is compatible with the Consistency principle (also in Nevins 2018), according to which, in Western South Slavic languages (and in Bantu), the single conjunct agreement is an option available only to gender, not to number. Such an assumption favours the view of gender and number as split features, with number feature operation settling before gender, hence number agreement is determined first. Moreover, each probe from a different participle is considered as a separate operation (also mentioned as a separate agreeing head in Bošković 2009: p. 473, Footnote 21). If it is the case that number and gender agreements do not have to target the same constituents, number and gender features are different bundles. Hence, we are talking about non-simultaneous procedures, if number and gender agreement do not follow the same pattern. Four different

⁴Important to note is that Aljović & Begović (2016), in their analysis, consider the agreement relations to be established in syntax proper, adopting van Koppen (2005), with post-syntactic filtering of vocabulary insertion. For detailed discussion, the reader is referred to Aljović & Begović (2016) and the analysis in Begović & Aljović (2015), preceding their conclusions in the subsequent publication.

possibilities are available: they can be split (sequential or simultaneous), or a single probe (sequential or simultaneous). Contra argument to single conjunct agreement not available with singular members (element with a singular number feature value) in BCMS is given in Arsenijević & Mitić (2016a but see also Mitić & Arsenijević 2019a in facilitating effect of plural agreement in number). If HCA in gender is available only in the plural, it will predict that HCA in gender will not happen if the highest conjunct is in singular, but Arsenijević & Mitić (2016a) show it does.

Assuming that the goals for each feature are different, and that &P only specifies the number, we expect agreement on the number with the whole coordination and agreement on the gender with a single member⁵. If gender and number are split bundles, or a single probe but in sequential order, the number is copied from &P and gender from the hierarchically/linearly closest conjunct (Peeking grammar in Marušić et al. 2015); though not controlled simply by the noun phrase closest to the agreeing predicate, but by the head of the last conjunct⁶, making it different than attraction errors. Hence, a verb can agree with the linearly closest conjunct on the surface, postulating a distributed approach to the closest conjunct agreement (CCA), as the result of the Peeking grammar (Marušić et al. 2015).

Default gender specification is assumed by the No-peeking grammar, where no percolation is present and both, number and gender, are copied onto the probes. In Marušić et al. (2015) and Nevins (2018), default agreement simply chooses a single gender value as the default when ordinary agreement ‘fails’, or when the conjunction head (&) is the controller. The difference between default and resolution lies in the two different values. As stated in Nevins (2018), ‘a resolution agreement, which selects a certain gender value from a pair of (non-)identical ones, is distinct from a default agreement, which is masculine’. The masculine feature is the least marked/specified and has the greatest probability of survival. For resolution to even be possible (as opposed to simply default or underspecified), the conjuncts must bear the appropriate number (Nevins 2018). Resolution in BCMS, a language with three genders, in conjunction with masculine and feminine/neuter, chooses masculine when resolving agreement⁷. When two feminine nouns are combined, the resolution is the feminine gender value; however, when neuter&neuter is used, there is no resolved neuter agreement and only the masculine value is seen. When a combination of genders includes masculine, the outcome is consistently masculine values. The masculine feature is difficult to tell whether it is default, resolution, or closest conjunct and therefore is avoided in this study. The feminine&neuter/neuter&feminine gender combinations behave as the most informative combinations.

In the following section (§3), we go into more details about the Peeking and No-peeking grammars. Before that, we discuss the last account to consider in this section, which is the one proposed by Aljović & Begović (2016). They offer an analysis according to which it is not the feature bundles but the exponents (realisation of the features) that are involved. The fourth double agreement

⁵ For more detailed discussion on differences in the nature of the two, see, e.g. Nevins (2018) using examples of Slovenian and BCMS, as well as Bantu language family.

⁶ Agreement relation is observed with the conjunct NP head and not any closest nominal, as in *Strunjače i korita, na kojima su spavali prasići, su bazdile/bazdila/*bazdili* ‘Mats-F.PL and troughs-N.PL, on which were sleeping pigs, stinked-F.PL/N.PL/*M.PL’.

⁷ But see the neuter default value in resolution as in Icelandic described, for example, in Corbett (1991). Nevins (2018) mentioned that the default gender is the one which is lexically specified on the conjunction head.

pattern of interest to be empirically explored in this work, HCA(CCA)+def, was previously introduced and analysed by Aljović & Begović (2016, HCA+def as they termed it⁸). Aljović & Begović (2016) argue that when conjoined noun phrases have different gender values, agreement relation of the modifier with HCA is acceptable. But the same condition does not act on the participle. Instead, we see the default masculine plural, which they label Plural Agreement. Following the DM framework, they propose a post-syntactic filtering constraint called the Vocabulary Insertion Filtering Hypothesis (VIFH). Under such proposal, the predicate is not subject to VIFH, but an attributive adjective is, resorting to the case licensing position (SpecT). This is the core difference between the two Probes and Vocabulary Items (VI) realising their features. Goal 1 (&P) can be underspecified for gender and will count as less specified. When it selects NP₁ (Goal 2) to agree with, it is because NP₁ counts as a more specified goal. After preliminary linearisation at PF, Vocabulary Items-insertion happens. Moreover, Begović & Aljović (2015) add that when selecting &P with which to agree (plural agreement in BCMS), the choice is motivated by some independent semantic requirement for a plural (adjective, wh-word).

Presence/absence of agreement with a coordinate wh-subject also serves as a piece of evidence for the reconstruction parameter in the following examples in (5) and (6) from English, mentioned in Bošković (1997: 87) and Kobayashi (2022: 82).

- (5) a. [A man and a woman] are/*is in the house.
 b. There is/*are [a man and a woman] in the house.
 c. There are/*is [four men and a woman] in the house.

A coordinated subject triggers plural agreement on a finite verb, as shown in (5a). If a subject remains in a postverbal position, as in (5b) and (5c), a finite verb only agrees with the first conjunct of the subject. If a coordinated subject is A'-moved, it triggers plural agreement (6) (Kobayashi: 83).

- (6) [Which man and which woman] do you think are in the house?

If the subject had to move directly to the A'-position, skipping the subject position (SPEC-T), the copular would show singular agreement. Plural agreement indicates that the wh-subject stops at SPEC-T and SPEC-C, creating two criterial configurations. As claimed by Kobayashi (2022), reconstruction allows wh-part to be interpreted at SPEC-C and the rest to be interpreted at SPEC-T.

2.3 *Hierarchy and adjacency*

In order to analyse a wide range of conjunct-sensitive agreement patterns, we utilise the distributed model. We want to determine if it can accurately represent grammatical derivations and eliminate ungrammatical examples. In the DM model, Arregi & Nevins (2012) propose linearisation as a post-syntactic process that introduces linear connections to the syntactic tree, without eliminating hierarchical relations. Murphy & Puškar (2018) advocate a post-syntactic mechanism for resolving

⁸Such a case was also found in Finnish, where the second position auxiliary shows FCA, but the participle that follows the conjoined subject shows plural agreement (see Crone 2016 mentioned in Nevins & Weisser 2018, and references therein).

features using impoverishment. The analysis of complementizer and verbal participle configuration was carried out by Citko (2018) and van Koppen (2005). According to Citko (2018), there are two distinct ways Agree can operate. One way employs a sequence of Single Agree operations, or one instance of Multiple Agree, with Multiple Agree yielding Resolved Agreement and Singular Agree requiring feature conflict resolution at PF, sensitive to proximity. Van Koppen (2005) exploits the idea of two equally local goals (adopted also in Begović & Aljović 2015) in accounting for complementizer agreement in some varieties of Dutch. Following the model proposed by van Koppen (2005), a Probe is allowed to agree simultaneously with both Goals and syntax creating two feature bundles on the Probe, which arrive at PF but only one gets realised.

3 The distributed model of conjunct agreement

Based on prior research, it has been observed that Western South Slavic languages exhibit agreement in number and gender in both pre-verbal and post-verbal contexts. As already explained, agreement with coordinated subjects presents some interesting problems for linguistic theory. One of the most interesting agreement patterns observed with coordinated subjects are cases of agreement with the closest conjunct, which have been rather successfully explained with the distributed model of agreement. Originating in DM, the distributed model assumes there is a component of the grammar that is relevant for derivation that is active even after the syntactic operations Merge, Move etc. are done constructing the syntactic structure. The model adopts a two-step Agree operation (Arregi & Nevins 2012), with the standard Agree operation split into Agree-Link and Agree-Copy, one establishing a link between the probe and the goal in syntax proper, and the other copying the feature values. When different feature values are found on conjuncts, the conflict can be resolved either by applying the default/resolution⁹ value or selecting one of the conjunct options, where the Probe must peek inside &P to find the Goal. In the following subsection, we discuss peeking and no peeking variants, and the two options in the peeking variant resulting from different timing of applying the Agree-Copy operation (either before or after linearisation of the syntactic structure). It is important to adopt the assumptions made in the proposal to accurately analyse the patterns in this study.

3.1 Peeking and No-peeking grammars

Marušič et al. (2015) address, among other things, whether the mechanisms of probe for number and gender features operate jointly or independently, as well as whether the alternatives are mutually exclusive for a language (cf. Bejar 2003 for person and number split features, each realising morphological endings with different syntactic objects of the same clause). If gender and number agreement are split, number can be copied from &P because number of the entire &P can be computed, but gender, unlike number, is not additive and thus &P does not always have a gender feature that is common for the two conjuncts, so gender feature has to be copied from one of the conjuncts (The peeking grammar). The choice of which conjunct is to agree with in the end depends

⁹As already mentioned above, in Footnote 2, the difference between the two terms might be obvious. See, for e.g., Slovenian data under disjunction (Marušič & Shen 2021), where the authors argue for different treatment and in favour of the resolution. We acknowledge that but will leave it on the side for now. See also Nevins (2018), arguing for two distinct agreement strategies, namely default vs. resolution.

on the type of prominence that is considered, and here the choice is either precedence or dominance¹⁰, as Marušič & Nevins (2010: p. 2, second paragraph) outlined, considering previous study on the last conjunct agreement in Slovenian, while briefly discussing head-initial and head-final languages (Marušič et al. 2007). The illustration of their proposal is repeated in Figure 2.

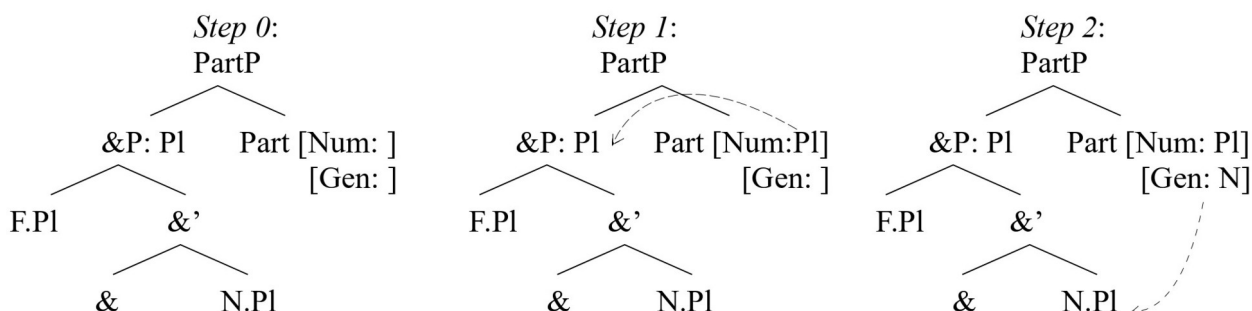


Figure 2: Deriving CCA (LCA) pre-verbally (Marušič et al. 2007: 37, p. 15).

Marušič et al. (2007) demonstrated that Slovenian does exhibit LCA, which cannot be analysed parallel to HCA. However, their treatment of LCA is attributed to genuine CCA in pre-verbal contexts. In the No-peeking grammar, the probe stops at &P, collects the necessary features and the operation is completed. As &P has no gender features, the gender feature value that is realised is the default masculine. Unlike the No-peeking grammar, the Peeking grammar considers the process by which the feature values of one of the conjuncts within the &P are copied to the probe(s). Gender feature values are copied from the hierarchically highest or first conjunct, resulting in HCA strategy, or from the linearly closest conjunct, yielding CCA.

In the remainder, we briefly go once again over the pre- and post-verbal structures before moving on to sandwiched structures. In the PreV order (SV) in BCMS, the subject-predicate agreement allows three agreement patterns: CCA, HCA, and def. LCA with NP₂ is CCA. The examples (3a-c) from introduction, repeated in (7a-c), illustrate this point.

CCA (LCA)

- (7) a. [*Nagrada* *i* *priznanja*] *su* *pripremljena*.
 award-F.PL and acknowledgments-N.PL AUX-PL prepared-N.PL
 ‘Awards and acknowledgments are prepared.’

HCA

- b. [*Nagrada i priznanja*] *su pripremljene.*
award-F.PL and acknowledgments-N.PL AUX-PL prepared-F.PL
'Awards and acknowledgments are prepared.'

def

¹⁰The treatment of deriving the same strategies in the syntax proper by Bošković (2009) is out of scope of the current study but I do acknowledge interesting aspects of it.

- c. [*Nagrade i priznanja*] *su pripremljeni*.
 award-F.PL and acknowledgments-N.PL AUX-PL prepared-M.PL
 ‘Awards and acknowledgments are prepared.’

In the PostV order (VS), HCA is present with the first conjunct (NP_i) post-verbally, making CCA, as demonstrated earlier (4a), repeated in (8a). LCA in (4b), repeated in (8b), is bad, while the default agreement pattern (4c), repeated in (8c), is degraded. According to my own grammatical judgment, as a native speaker, LCA in PostV is ungrammatical and def PostV degraded. This assessment is consistent with the experimental findings of Willer-Gold et al. (2016).

CCA (HCA)

- (8) a. *Pripremljene su [nagrade i priznanja]*.
 prepared-F.PL AUX-PL award-F.PL and acknowledgments-N.PL
 ‘Awards and acknowledgments are prepared.’

*LCA

- b. **Pripremljena su [nagrade i priznanja]*.
 prepared-N.PL AUX-PL award-F.PL and acknowledgments-N.PL
 ‘Awards and acknowledgments are prepared.’

??def

- c. *??Pripremljeni su [nagrade i priznanja]*.
 prepared-M.PL AUX-PL award-F.PL and acknowledgments-N.PL
 ‘Awards and acknowledgments are prepared.’

As shown in (7a-c) and (8a-c), HCA is possible in both SV and VS (CCA). LCA is good pre-verbally (CCA) but ungrammatical post-verbally. The default agreement pattern is judged grammatical PreV but degraded PostV, in line with Willer-Gold et al. (2016) but contra Murphy and Puškar (2018). The patterns are summarised in Table 3.

Pattern	PreV (SV)	PostV (VS)
CCA	✓	✓
HCA	✓	✓
LCA	✓	*
def	✓	??

Table 3: Attested agreement patterns in BCMS in previous studies.

3.2 Two probes agreeing with different/same goal(s): sandwiched contexts

When we look at sandwiched setups, the two participles (agreeing probes) on opposite sides of coordination don't always agree with a single/same conjunct(s) when marking gender. Instead, BCMS displays multiple agreement patterns, like Slovenian (9).

- (9) Slovenian (Marušič & Nevins 2020: 18)

Na sejmu so bile [krave in teleta] prodana/prodane/prodani.
 on fair AUX-PL been-F.PL COW-F.PL and calf-N.PL sold-N/F/M.PL
 ‘Cows and calves were sold on the fair.’

Three agreement patterns shown in (9) are: (i) HCA, agreement with the hierarchically highest conjunct (NP₁) on both Participles, (ii) CCA, the closest conjunct agreement, where the first verbal element agrees with HCA (NP₁) and the second verbal element agrees with LCA (NP₂), and (iii) HCA+def¹¹, where the first verbal element agrees with the highest and closest conjunct (NP₁) and the second participle exhibits default agreement (&P). But there are considerably more theoretically possible patterns. Following Marušič & Nevins (2020), there are 9 theoretically possible agreement patterns in sandwiched configurations. Not all of the theoretically possible patterns are grammatical. According to my assessments, three out of nine patterns, presented in (10a-c), are grammatical¹² in BCMS.

Double CCA (HCA+LCA)

- (10) a. *Za danas su bile [nagrade i priznanja] pripremljena.*
 for today AUX-PL were-F.PL award-F.PL and acknowledgments-N.PL prepared-N.PL
 ‘Awards and acknowledgments were prepared for today.’

Double HCA (CCA+HCA)

- b. *Za danas su bile [nagrade i priznanja] pripremljene.*
 for today AUX-PL were-F.PL award-F.PL and acknowledgments-N.PL prepared-F.PL
 ‘Awards and acknowledgments were prepared for today.’

HCA (CCA)+def

- c. *Za danas su bile [nagrade i priznanja] pripremljeni.*
 for today AUX-PL were-F.PL award-F.PL and acknowledgments-N.PL prepared-M.PL
 ‘Awards and acknowledgments were prepared for today.’

The double default strategy (11a) is found by me degraded, or to its best very marginally acceptable, while other five double agreement patterns ungrammatical.

Double def

- (11) a. *??Za danas su bili [nagrade i priznanja] pripremljeni.*
 for today AUX-PL were-M.PL award-F.PL and acknowledgments-N.PL prepared-M.PL
 ‘Awards and acknowledgments were prepared for today.’

¹¹ Marušič & Nevins (2020: Footnote 5) take the pattern HCA+def to be judged poorly.

¹² The author leaves open the possibility that not all native speakers of BCMS share preferences for the above grammaticality judgments of the double agreement patterns to the same extent, and that is precisely the purpose for conducting the experimentally-controlled acceptability experiment.

Double LCA

- b. **Za danas su bila [nagrade i priznanja]* *pripremljena.*
 for today AUX-PL were-N.PL award-F.PL and acknowledgments-N.PL prepared-N.PL
 ‘Awards and acknowledgments were prepared for today.’

def+LCA

- c. **Za danas su bili [nagrade i priznanja]* *pripremljena.*
 for today AUX-PL were-M.PL award-F.PL and acknowledgments-N.PL prepared-N.PL
 ‘Awards and acknowledgments were prepared for today.’

def+HCA

- d. **Za danas su bili [nagrade i priznanja]* *pripremljene.*
 for today AUX-PL were-M.PL award-F.PL and acknowledgments-N.PL prepared-F.PL
 ‘Awards and acknowledgments were prepared for today.’

LCA+HCA

- e. **Za danas su bila [nagrade i priznanja]* *pripremljene.*
 for today AUX-PL were-N.PL award-F.PL and acknowledgments-N.PL prepared-F.PL
 ‘Awards and acknowledgments were prepared for today.’

LCA+def

- f. **Za danas su bila [nagrade i priznanja]* *pripremljeni.*
 for today AUX-PL were-N.PL award-F.PL and acknowledgments-N.PL prepared-M.PL
 ‘Awards and acknowledgments were prepared for today.’

See Table 4 for a breakdown of the patterns shown in (10a-c) and (11a-f).

Pattern	Configuration	Grammaticality
Double CCA(HCA+LCA)	V _F FN V _N	✓
Double HCA(CCA+HCA)	V _F FN V _F	✓
HCA(CCA)+def	V _F FN V _M	✓
Double def	V _M FN V _M	??
Double LCA	V _N FN V _N	*
def+LCA	V _M FN V _N	*
def+HCA	V _M FN V _F	*
LCA+HCA	V _N FN V _F	*
LCA+def	V _N FN V _M	*

Table 4: Sandwiched agreement strategies in BCMS.

Our contribution includes the demonstration that LCA is grammatical only if it is linearly nearest conjunct, that is, genuine CCA after the highest conjunct has been linearised. In the configurations analysed in this study, it is observed that distant LCA (referred to as DCA in Marušič et al. 2007 and subsequent studies of theirs) is not viable because it is never the closest hierarchically and therefore does not establish a match in the first part, the Agree-Link. On the other hand, HCA, as the distant conjunct pre-verbally, remains hierarchically highest/closest (Marušič et al. 2015). It is worth noting that HCA and LCA are not treated as parallel strategies (in line with Marušič et al. 2007). The last conjunct is not hierarchically highest or closest to either of the participles in sandwiched configurations, nor it is linearly closest to the first linearised participle. It is only the closest to the second participle, which adheres to CCA in pre-verbal contexts. In the upcoming section (§4), we lay out the hypotheses and predictions for the proposed underlying mechanisms that ideally render grammatical patterns (10a-c) and exclude the ungrammatical ones (11a-f).

4 Hypotheses and predictions

As outlined in the introduction section, the central question of the paper is what happens if different goals with distinct gender features, as well as a single goal, are most accessible to two verbal probes of the same clause, and how to account for such agreement patterns. We hypothesise that two participles are modelled as if two probes act independently, and each Part having split gender and number probes. In order to test it experimentally, the following generalisation serves as experimental hypothesis in the current study: Participles in sandwiched configurations can agree (i) each with the closest conjunct, (ii) each with the hierarchically highest conjunct, and (iii) can display the default agreement pattern, regardless of the conjuncts' gender, for the second Part but not for the first Part.

Next, we set up predictions. Double CCA(HCA+LCA) and HCA(CCA)+def patterns are predicted to be judged as acceptable as CCA/def PreV and HCA(CCA) in PostV orders, respectively. The double HCA(CCA+HCA) pattern is predicted to be judged as acceptable as HCA(CCA) PreV and PostV. Alternatively, we find that different double agreement patterns do not have strong effect on the acceptability (H_0), with respect to the data. The following agreement strategies are predicted from the above hypotheses:

- i. If participles act as independent probes agreeing with linearly closest conjuncts in sandwiched configurations, then double CCA(HCA+LCA) is judged acceptable by BCMS speakers;
- ii. If participles act as independent probes displaying single conjunct agreement, then double HCA(CCA+HCA) is judged acceptable by BCMS speakers;
- iii. If Part's are independent probes, HCA(CCA) on the first Part is good and default on the second Part is good, yielding the HCA(CCA)+def strategy predicted as acceptable by BCMS native speakers;
- iv. If participles display the default conjunct agreement for the second but not for the first Part, then the double default agreement pattern is to be judged as bad by BCMS speakers.

Each prediction reflects three double agreement patterns judged above as acceptable and the fourth one judged as unacceptable. As an anonymous reviewer pointed out, double HCA yields the same pattern as CCA+HCA, and HCA+def same as CCA+def. We acknowledge that and thank them for the observation, it is a good point. We use the notation double HCA and HCA+def purely for the sake of simplicity and consistency with previous terminology, as we adopted the names from Marušič and

Nevins (2020). Ideally, we want other agreement patterns negatively judged by me to be experimentally confirmed, as the proposed model should ideally account exactly for the good ones and rules out the bad ones.

4.1 Ungrammaticality of distant NP₂ (LCA)

We saw that not all logically possible combinations are grammatical. We ask then what restricts the order and combinations of grammatical patterns. Unattainable double agreement patterns with the last conjunct in sandwiched contexts come down to the ungrammaticality of LCA with postverbal subjects. While Agree-Link establishes agreement between a verbal element and &P, Agree-Copy never copies the values for gender from the hierarchically lower and linearly more distant conjunct. In other words, the last conjunct is predicted not to control agreement post-verbally. All patterns that include LCA on the first verbal element (reflecting post-verbal order) are predicted to be ungrammatical. This includes double LCA, LCA+HCA (the reverse of double CCA), and LCA+def.

Unlike distant NP₂, distant NP₁ conjunct is a valid agreement controller. The reason is that NP₁ is either hierarchically or linearly closest, while NP₂ can be only linearly closest when pre-verbal. Pre-verbal NP₁ is linearly distant but hierarchically the highest and closest conjunct, and post-verbal NP₁ is hierarchically and linearly the closest conjunct. Since PreV HCA and PostV HCA(CCA) are grammatical (experimentally attested as robust in previous studies cross-linguistically), the double HCA(CCA+HCA) pattern in sandwiched context is predicted to be grammatical.

The last three double agreement patterns are predicted to be bad for additional reasons. These are double def, def+CCA(LCA), and def+HCA. PostV def was found to be degraded in Gold-Willer et al. (2016), as well as according to my judgments. To start with the default agreement pattern in sandwiched configurations on the first Part (reflecting PostV def) will be not preferable. Hence, no double agreement patterns starting with PostV def are expected. But since pre-verbal default is attested as good, in sandwiched configurations HCA(CCA) followed by def is to be good.

4.2 Interim summary

To summarise what we discussed up to now, we observe the following situations: two probes (Parts) on each side of the coordinated subject independently probe and get the gender features from the same goal or from two different goals. Also, each Part probes separately for number and gender feature (split probe). For now, we investigate and keep our attention on gender, with constant plural number and gender combination F+N. The following agreement patterns are predicted to be attested as grammatical: double CCA, double HCA, and HCA+def. To test these predictions, we performed an experiment, presented in the next section (§5).

5 Acceptability rating experiment

5.1 Method and materials

The experiment is a conceptual replication of Marušič & Nevins (2020). By manipulating different agreement strategies in participial sandwiched configurations, we want experimentally to confirm or dispute the hypothesis and predictions regarding BCMS data outlined in the previous section. The initial pilot study with 11 participants was constructed using the online software PC Ibex (Drummond 2011) and helped us refine the methods and hypotheses. The upgraded experiment from which all the

quantitative data are collected for this study involves an acceptability rating paradigm on the Likert scale, created and hosted by the web-based Gorilla Experiment Builder (Anwyl-Irvine et al. 2018).

5.1.1 Design

Four double agreement patterns were designed as a factor (agreement patterns) with levels. Manipulated agreement strategies were double CCA, double HCA, HCA+def, and double def. The dependant variable was acceptability judgement rate on the Likert scale, where 1 meant completely unacceptable and 7 absolutely acceptable. Control conditions were used to establish the baseline, as well as for comparisons to the critical sandwiched strategies. Sentences with control and critical items consisted of the following constant variables: (i) coordinated nouns F+N gender combination across all conditions, (ii) plural (PL) number feature value across all conditions, (iii) all nouns were inanimate and no mass nouns were included, (iv) participial sandwiched configurations of pluperfect tense in passive voice in BCMS, and (v) all sentences were declarative.

Out of six double agreement patterns, judged as ungrammatical in the previous section, we decided to include two in the experiment: double def as a critical experimental condition, predicted as bad (hence the other negatively judged patterns involving it will be confirmed as bad) and double LCA as a negative control, already seen as bad robustly across studies and languages (hence all the other double agreement patterns involving it will be confirmed as bad).

We had two positive controls and two negative controls. Positive controls included a sandwiched simple configuration (non-coordinated subject) and the preverbal default pattern with a coordinate structure, PreV def. We chose negative controls to be PostV def, attested as degraded previously in Slovenian but controversial in BCMS, and double LCA. The total of eight conditions, critical and control agreement patterns, are presented in Table 5.

Agreement pattern	Configuration
1. Double CCA	V _F FN V _N
2. Double HCA	V _F FN V _F
3. HCA+def	V _F FN V _M
4. Double def	V _M FN V _M
5. Control Sand. Simple	V _F F V _F
6. Control PreV def	FN V _M
7. Control PostV def	V _M FN
8. Control Double LCA	V _N FN V _N

Table 5: Control and critical conditions.

5.1.2 Stimuli

Experimental stimuli included declarative affirmative sentences of coordinate subjects placed between participles, that were controlled for gender and number features, namely F.PL+N.PL. Coordination type under investigation involved only conjunction, plural number feature, and

inanimate nominals¹³. The stimuli were presented visually to participants. Sentence materials were collected from different earlier studies in BCMS on conjunct agreement and adjusted accordingly. Experimental items are listed in Appendix 1. Stimuli examples illustrating critical conditions are presented in (12).

(12) Sample of material for critical conditions in the Acceptability rating task.

Condition 1 (double CCA):	<i>Na izvor su bile košarice i pletiva donesena.</i>
	on spring AUX-PL were-F.PL basket-F.PL and knitwear-N.PL brought-N.PL
Condition 2 (double HCA):	<i>Za petak su bile nagrade i priznanja planirane.</i>
	for friday AUX-PL were-F.PL award-F.PL and acknowledgment-N.PL planned-F.PL
Condition 3 (HCA+def):	<i>Jedino su bile obale i mora ostali u lijepom sjećanju.</i>
	only AUX-PL were-F.PL coast-F.PL and sea-N.PL stay-M.PL in nice memory.
Condition 4 (double def):	<i>Tamo su bili kuverte i pisma stajali.</i>
	there AUX-PL were-M.PL envelop-F.PL and letter-N.PL stand-M.PL

Eight (8) conditions (four critical and four control) and ten (10) sentences per condition make eighty (80) items in total. Trials were randomised in the way that they each appeared only once in the whole experiment session, and the same condition with different lexical choice is not repeated more than twice in a consecutive order. Each participant saw all the items in a random order. In order to maximise the statistical power, we calculated power analysis (Westfall et al. 2014) with the effect size $d = 0.3$, fifty-six ($n = 55$) participants, and obtained the power of 0.819. Potential confounds we considered and features kept equal across the critical stimuli are the following:

- (a) sentence length ranging from 5-10 words, 18 syllables on average;
- (b) declarative affirmative sentences used: no question forms or negation used, to avoid confounds of syntactic/semantic type, as well as possibility of rating positive vs. negative connotations;
- (c) stimuli presented in visual modality: in order to avoid competition between modalities (visual (reading) vs. auditory (listening));
- (d) syntactic structure with coordinated nominals of F.PL+N.PL between two finite-verb forms: past tense of be + L participle (active past participle), and past tense of be + N(T) participle (passive past participle or adjectival participle);
- (e) minifying structural complexity within &P in the *Nominative* case: no modifiers (quantifiers/demonstratives/numbers/adjectives) used, only bare noun phrases, also no hierarchically nested phrases used;
- (f) mono-clausal sentences only used, no clausal subordination/embedded clauses;
- (g) agentivity: following semantic theta roles, nominals were bearing the patient thematic role;
- (h) argument structure: keeping constant argument structure of verbs in passive voice (processing demands of passive constructions acknowledged) where patients are subjects. External argument slot, SpecTP (grammatical role of a subject position) occupied by nominals with

¹³Configurations such as Adj+V and secondary predication, as well as adding disjunction, animate nouns, and paucal number, left for future studies.

grammatical role of objects (like unaccusative structures whose grammatical subjects are not semantic agents);

- (i) perfectivity: two verbs of imperfective aspect used ('*stajati*' and '*planirati*'), the rest all perfectives;
- (j) predicates involved: no nominal, but only verbal and adjectival predicates used of intention, state, and achievement; no collective nor reciprocal verbs in predicates used;
- (k) imageability: concrete nouns only used, no abstract nor de-verbalized nouns used;
- (l) animacy: nominal items in coordination kept inanimate across conditions.

5.1.3 Fillers

Apart from eighty (n = 80) experimental trials, forty (n = 40) fillers were used as distractors. Filler sentences used in the experiment were taken from Bosnian, Croatian, Montenegrin, and Serbian sources, ten sentences per language variety, and customised for the grammatical and ungrammatical conditions (regarding word order and argument structure). Filler items are listed in Appendix 2. The stimuli were randomised between experimental and filler items in the main experiment procedure.

5.2 Participants

Sixty-two (n = 62) participants completed the experiment, with sixty (n = 60) self-reported native speakers of BCMS. Fifty-six (n = 56) persons were included in the analysis (44 female), with mean age of 36 (age range 19-71, SD = 12.3). Criteria for the analysis inclusion was 100% accuracy to all attention tasks, as well as 100% accuracy to all ungrammatical filler items. Non-linguist and linguist study participants came from BCMS natives of different variants/dialect regions. A demographic question was included in the pre-experiment survey, as well as the question whether participants had academic background in linguistics or related discipline. Though, data sample is collected based on colloquial language and language region was not controlled for¹⁴. All informants reported normal/corrected to normal sight. No participant reported speaking or writing disorders. An informed consent was provided in electronic form before the experiment (ticked by the checked box). Informants were unaware of the purpose of the research and the research question motivation. The recruitment of participants was carried out via Prolific, Amazon Mechanical Turk (M Turk), and Facebook social platform. Twenty-eight (n = 28) informants were recruited on the Prolific platform, for a compensation of £1.6 per participant. Twenty-six (n = 26) persons were recruited on Facebook and voluntarily completed the survey for no compensation (Facebook pages are ordered as the request was posted, *Anketalica* 'Questionnaire', *Kako biste vi rekli* 'How would you say it', *Popunjavanje upitnika za istraživanja* 'Filling out research questionnaires', and *Prevodilac* 'Translator'). Eight (n = 8) persons were recruited on M Turk, where each participant earned \$1.5 per session.

5.3 Procedure

Acceptability judgments of agreement patterns were measured on a rating scale, asking a simple question: 'How acceptable is this sentence to you?' The scale went from 1 (completely unacceptable) to 7 (completely acceptable). Before the actual experiment started, informants were trained in the use

¹⁴ Australia (2), Bosnia and Herzegovina (7), Canada (1), Croatia (19), Germany (1), Serbia (23), Slovenia (1), USA (1), and NA answers (7).

of a rating procedure in a sentence comprehension task. Subjects read/saw a sentence with simultaneously presented 7-point Likert scale below. They rated how natural a sentence was to them before proceeding to the next one. Practice block comprised ten sentences (five grammatical and five ungrammatical, randomised). Following the practice block, an informant was presented with the real experiment trials in the same manner. Each sentence was individually presented on the screen with a rating scale, one after another in random order. Once the participant gave the grade, they proceed to the next example by clicking the button "next". Attention task (total of 7), "This is attention task. Give rate of ___.", translated to BCMS, was randomly performed among experimental and filler items in the experiment to check attentiveness.

Subjects were tested individually in front of their computer/tablet/phone screen (experiment accessible via online connection on computer, tablet, or smartphone). The time allowed for the rating was not limited, however informants were instructed to complete the task as fast as possible, and informed that the total session takes approximately 15-20 minutes to complete. In the experiment builder, 30 minutes was allocated as the maximum time for the experiment to be completed. The experiment consisted of a single session, took approximately 15-20 minutes for completion, and was carried out in accordance with the existing international regulations concerning ethics in research. A small correction was done after 17 people already took the experiment (the middle of the rating scale (4) was not visible on the screen since the very start but only when participants touched the screen). Settings features were adjusted but it did not affect the quality of ratings or misinterpretation of the procedure by participants, as no difference is observed in the results between these and the following subjects (the correct ratings of ungrammatical and filler sentences were checked by the experiment builder, a BCMS native speaker).

5.4 Data analysis

Acceptability rate response as a dependent variable was adjusted as a function on agreement patterns as conditions of interest (independent variable) with levels. Significance test for a multilevel predictor was performed, using one-way analysis of variance (ANOVA). The one-way ANOVA test results indicated there was an overall significant difference (Appendix 3). Due to the nature of the data, analysis of variance included Kruskal-Wallis test, performed in RStudio (RStudio Team 2019), followed by Dunn test for multiple pairwise-comparison between groups.

Regression analysis included mixed effects linear regression, using lme4 package (Bates et al. 2015), to estimate the model with double agreement patterns as categorical predictors, standardised ratings as a dependent variable, and by-participant varying intercept and varying slope, with slope/intercept correlation. We interpreted the model in terms of the predictions and discussed it with regards to expressing our main questions and hypotheses in the study. More on that in Section 6, where we present results and statistical analyses, followed by discussion.

6 Results and discussion

Results data comprise 560 observations per condition (56 participantsX10 items), across 8 different conditions (total 4480). Four critical conditions (56X40) comprise 2240 data points. Summary statistics report the mean, median, standard deviation, and standard error of experimental conditions: $M = 4.58$, median = 5, $SD = 2.08$, $SE = 0.03$. Average ratings of eight agreement patterns, in

descending order, are summarised in Table 6 and graphically presented in the bar chart (Figure 3), with total mean 4.58 (dashed red line).

Pattern	Configuration	Average Rating
PreV def	FN V_M	5.62
Sand. Simpl	V_F F V_F	5.14
Double CCA	V_F FN V_N	4.67
PostV def	V_M FN	4.53
HCA+def	V_F FN V_M	4.46
Double HCA	V_F FN V_F	4.24
Double def	V_M FN V_M	4.21
Double LCA	V_N FN V_N	3.82

Table 6: Average results per condition.

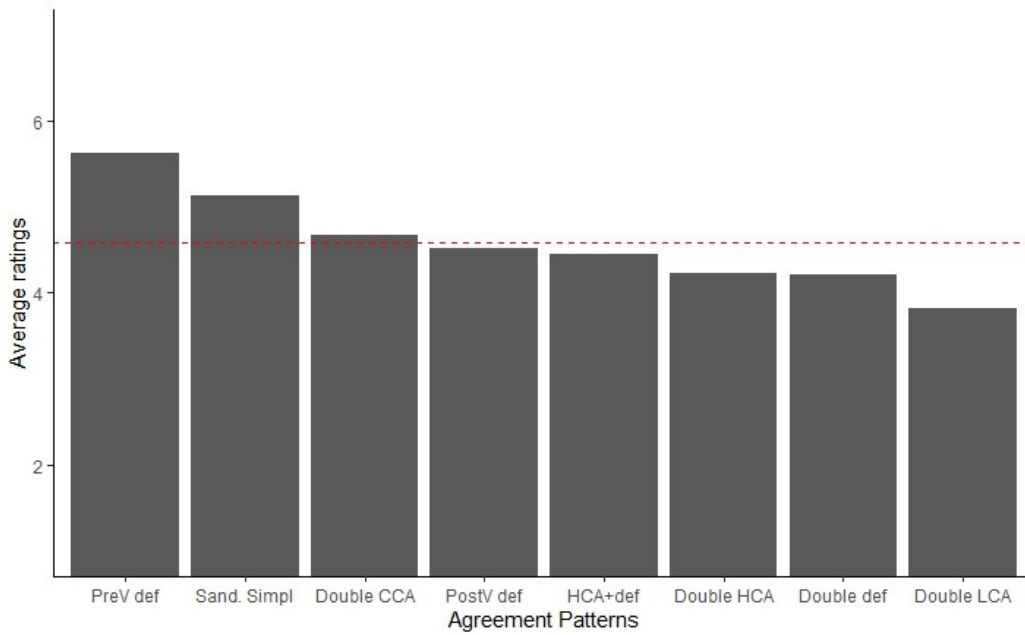


Figure 3: Bar chart of average ratings.

All positive control groups showed ceiling acceptance rates. The highest was PreV def ($M = 5.62$, $SD = 1.73$), followed by Sandwiched Simple ($M = 5.14$, $SD = 1.92$), attested as empirically robust pattern among BCMS speakers. The negative predicted control, PostV def ($M = 4.53$ and $SD = 2.15$), was in the third place amongst the controls and in the fourth place in total. The other negative control pattern, double LCA, had the lowest average acceptance rate ($M = 3.82$ and $SD = 2.14$). Boxplot is presented in Figure 4, with the mean (red rhomboid) and median line, in descending order, and total mean 4.58.

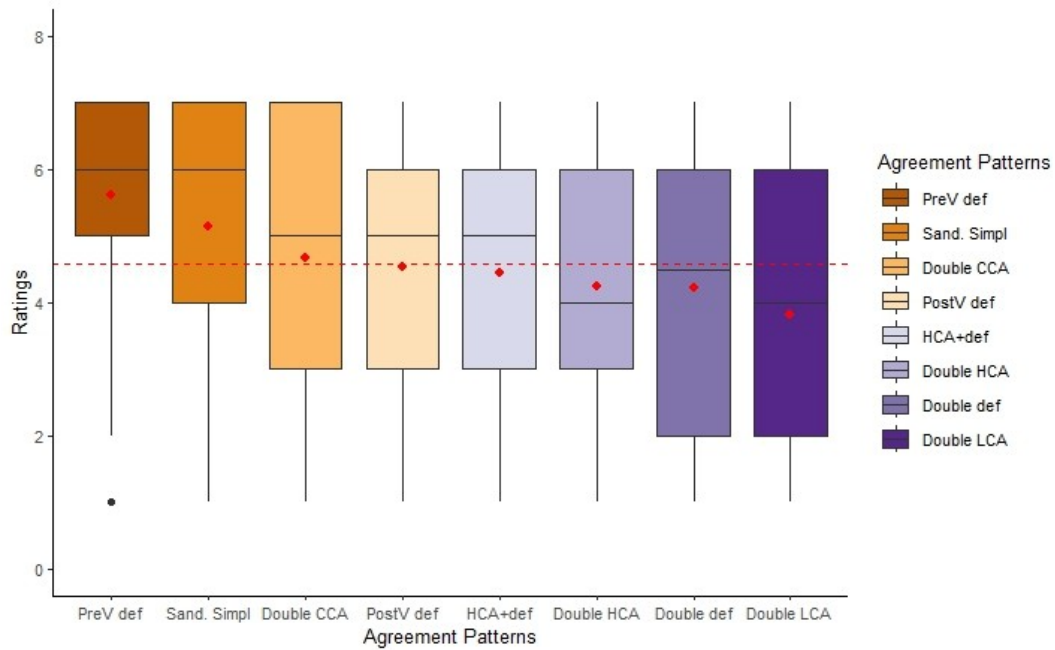


Figure 4: Boxplot of raw acceptance ratings.

Descriptive statistics for critical, four double agreement patterns, are $M = 4.39$, median = 5, $SD = 2.04$, $SE = 0.04$. The average ratings of double agreement patterns have $M = 4.39$, median = 4.35, $SD = 0.21$, $SE = 0.1$. The average rating of double CCA was the highest ($M = 4.67$, $SD = 1.99$), higher than average rates of HCA+def ($M = 4.46$, $SD = 2.04$), double HCA ($M = 4.24$, $SD = 1.98$), and double def ($M = 4.21$, $SD = 2.08$). Average ratings of critical conditions in descending order are summarised in Figure 5 with total mean 4.39 (dashed red line), with boxplot in Figure 6.

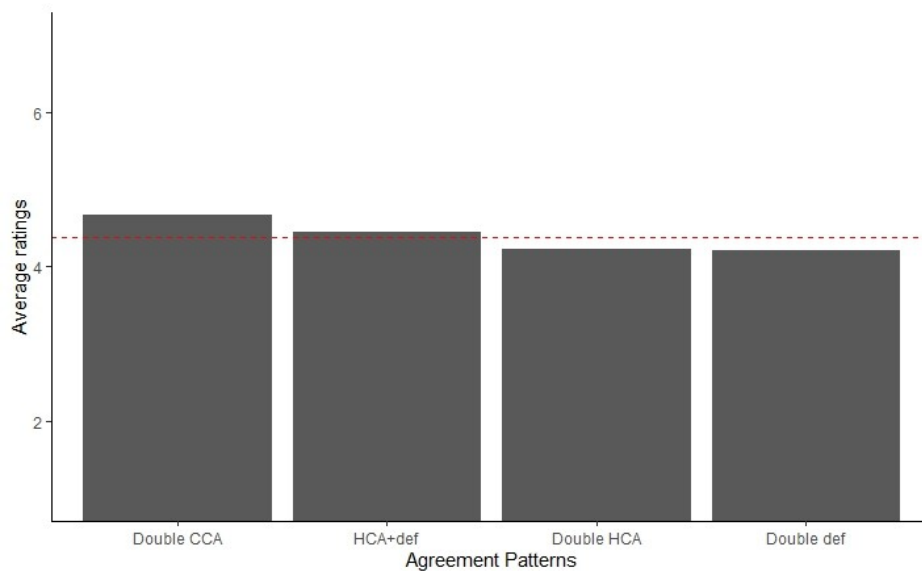


Figure 5: Average ratings of critical patterns.

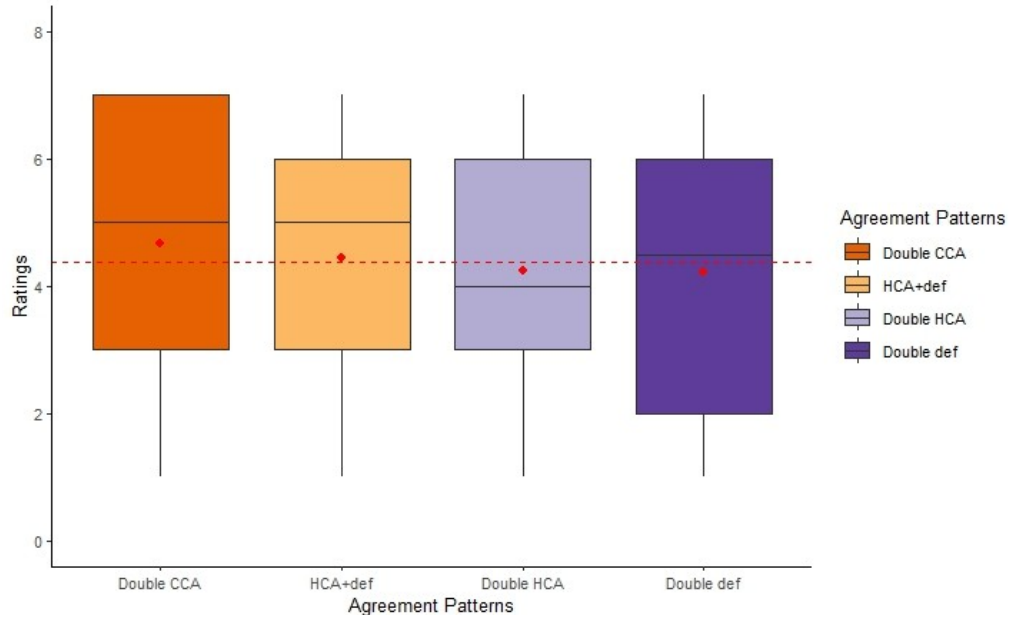


Figure 6: Raw ratings by double patterns.

Density graph (Figure 7) of double agreement patterns' ratings helped us understand how the data are distributed. Due to the negative skewness of data, Kruskal-Wallis test was performed. The results showed a statistically significant result (see Appendix 4). Post-hoc Dunn test performed multiple pairwise-comparison between groups, with Bonferroni correction for p values (Appendix 5). Results show no significant difference between double HCA and double def ($p = 1.0$), nor between double HCA and HCA+def ($p = 0.16$). Significance was not present neither between HCA+def and double def ($p = 0.14$), nor between HCA+def and Double CCA ($p = 0.2$). The last pairwise comparison, HCA+def and double CCA, confirms the validity of the analysis of Marušič and Nevins (2020: Footnote 5), where HCA+def in Slovenian is showed to be ungrammatical, as opposed to BCMS. Difference was significant between double CCA and double HCA ($p = 0.0005$), as well as between double CCA and double def ($p = 0.0005$).

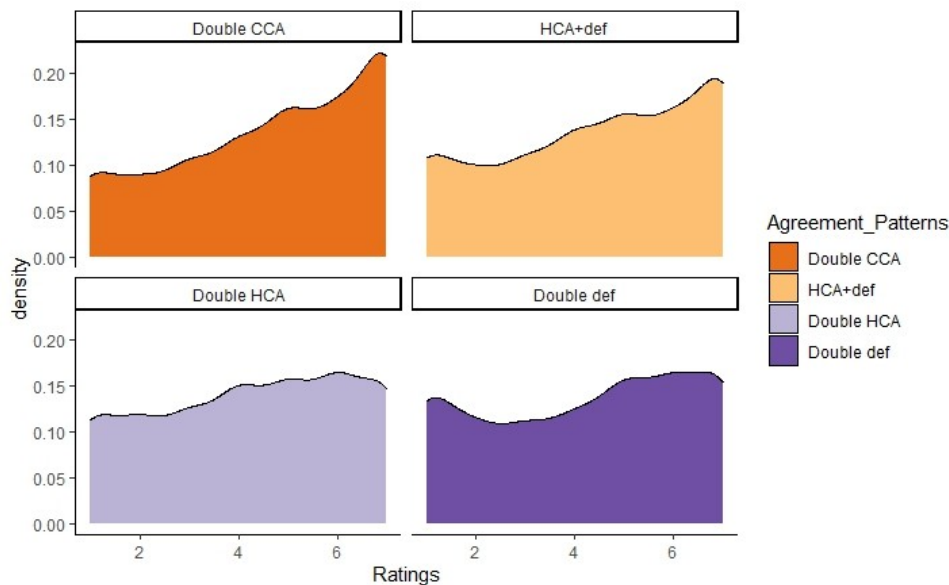


Figure 7: Density graph.

Regarding the agreement patterns double HCA and double def, even though they have close average ratings and are not significantly different, the counts in their ratings are drastically different if we employ visual inspection of the rating counts (Appendix 3, Table 2) and density graph in Figure 7. In the case of double def, ratings are more concentrated on two extreme values, 1 and 7, with much less middle ratings, implying that speakers rate the pattern as two extends, completely acceptable or unacceptable, and not as 'less good but still acceptable' closer to the mid ratings. On the other hand, double HCA received around 20 more middle ratings (4), and less extreme rating values (1 and 7), and the mean value is dragged higher than the median (Figures 4 and 6). Hence, we grouped the two agreement patterns as different, with double def as unacceptable. We elaborate on it in the next subsection, discussing the regression model fitting the data, and confirming the predicted unacceptability of double def and acceptability of double HCA.

Introducing individual differences, data distribution relevant for our analysis includes plotting categorical differences of acceptability ratings considering differences by participants. In Figure 8 below we compare distributions of raw agreement patterns' ratings by participants as a factor on the acceptability rating effect of double agreement patterns, in increasing order. In the plot we see that while some participants' agreement ratings are mostly in top half or around mid, others have them spread. Though very few, only four (41, 50, 60, 61) to be precise, have them very low in bottom half.

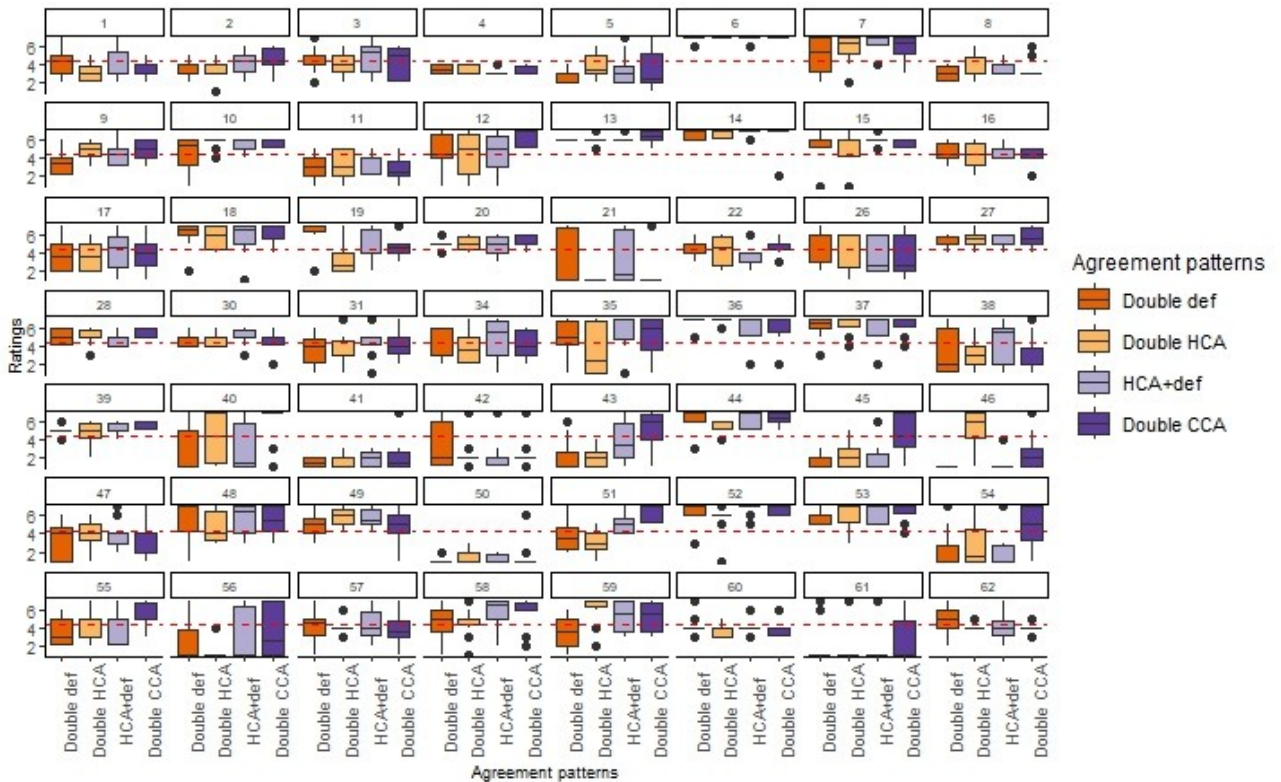


Figure 8: Distribution plot of double agreement patterns' raw rates by participants (mean, 4.39, dashed red line).

Nevertheless, average scores by double agreement patterns show much better distribution. That is the reason why we not only needed to "centre" categorical predictor levels (agreement patterns) by sum-coding them, but it was needed as well to standardise, or z-score, the response ratings.

Figure 9 depicts the total mean (red dashed line) and each agreement patterns' means (red rhomboid) of standardised rating scores (z-scores) by participants (each participant contributes one data point, an average per pattern), in increasing order (summary statistics of z-scores in Appendix 6, Table 3). What follows is fitting a model, covered in the next subsection.

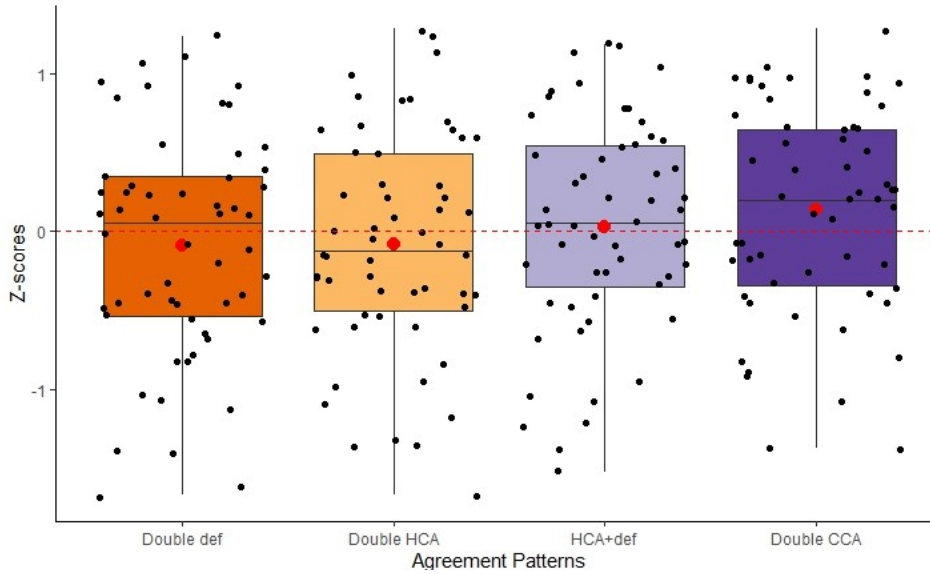


Figure 9: Mean z-scores on double agreement patterns with data points.

6.1 Significance testing in the regression context: Mixed effects linear regression

Regression analysis was conducted to examine the effect of double agreement patterns (the predictor) on acceptability rating (the outcome). If there are multiple data points for the same agreement pattern, this introduces a dependency into data set that needs to be modelled. As we want to measure individual differences for the effect, we consider random effects (with random intercepts and random slopes), using mixed modelling. A regression analysis was performed to fit data to a linear model, computing and plotting confidence intervals and standard errors for regression coefficients, and communicating uncertainty for the predictors.

In the analysis, we created a model to fit the data ($\text{Ratings} \sim \text{Agreement Patterns} + (1 + \text{Agreement Patterns} \mid \text{Participant})$). We standardised result rating scores into z-scores, with the scale function in R studio. As already mentioned, for categorical factor levels, sum contrasting was used and we have effectively “mean-centred” all categorical variables. By using sum contrasts we switch from having estimates that reflect marginal effects to estimates that reflect main effects. In such cases, the intercept reflects the estimate when all conditions equal 0. For sum contrast coded variables, this refers to the mid-point between the categories. As such, the intercept in this regression model that uses sum contrasts becomes the grand mean of all of the cells (z-scored). The model summary is outlined in Table 7, showing statistically significant differences between the reference level (intercept) and agreement pattern factor levels.

The fit of the model (R^2), as part of the model summary in Table 7, shows a moderate positive correlation, accounting for the 42% of variance in the dependant variable, and confirming that the regression model fits relatively well the original data. A graphical display of a coefficient table is shown on the coefficient plot of the model (Figure 10), with each coefficient corresponding 95% confidence interval. Visualising the fixed effects' estimates previously outlined in the model summary in Table 7, the plot in Figure 10 depicts the model's coefficient estimates.

Acceptability Ratings					
MODEL INFO:					
Observations: 2240					
Dependent Variable: scale (Ratings)					
Type: Mixed effects linear regression					
MODEL FIT:					
AIC = 5448.89, BIC = 5534.60					
Pseudo-R ² (fixed effects) = 0.01					
Pseudo-R ² (total) = 0.42					
FIXED EFFECTS:					
	Est.	S. E	t val.	d.f.	p
(Intercept)	0.00	0.08	0.00	56.01	1.00
Double CCA	0.14***	0.04	3.42	56.81	0.001***
HCA+def	0.03	0.03	0.99	130.57	0.32
Double HCA	-0.08	0.04	-1.85	56.02	0.07
Double def	-0.09*	0.04	-2.33	61.74	0.02*
<i>p values calculated using Satterthwaite d.f.</i>					
*** <i>p</i> < 0.001; ** <i>p</i> < 0.01; * <i>p</i> < 0.05.					
RANDOM EFFECTS:					
Group	Parameter	Std. Dev.			
Participant	(Intercept)	0.61			
Participant	Double CCA	0.22			
Participant	HCA+def	0.09			
Participant	Double HCA	0.24			
Participant	Double def	0.20			
Residual		0.76			
Grouping variables:					
Group	# groups	ICC			
Participant	56	0.39			

Table 7: Model summary.

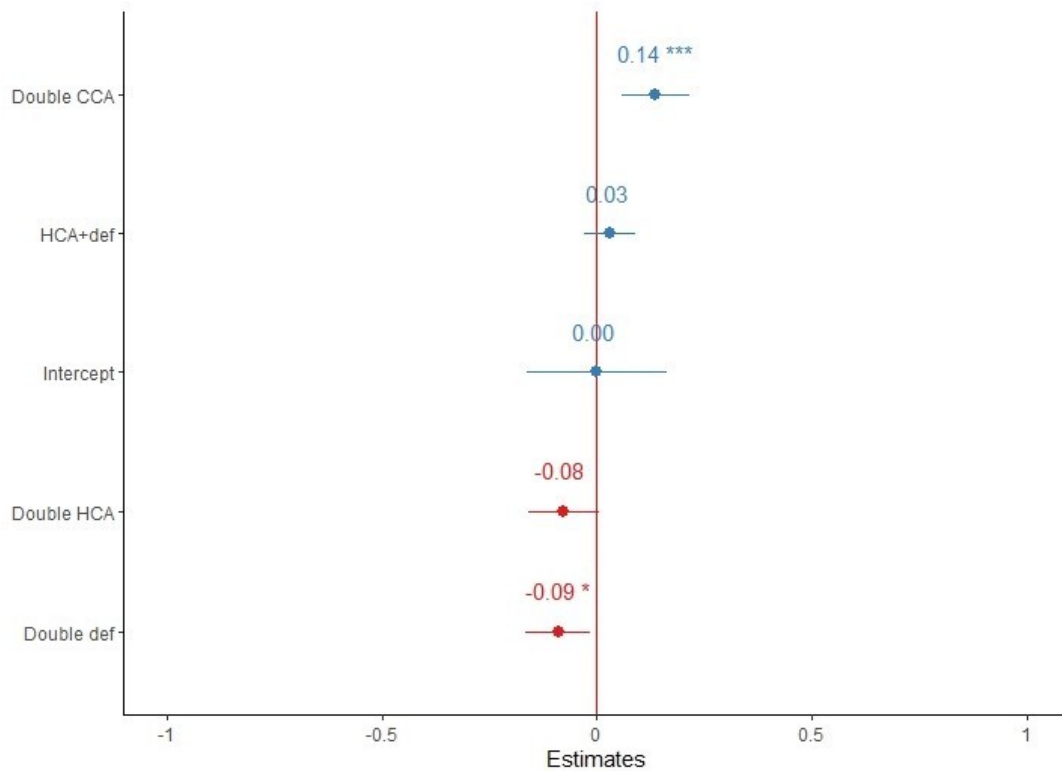
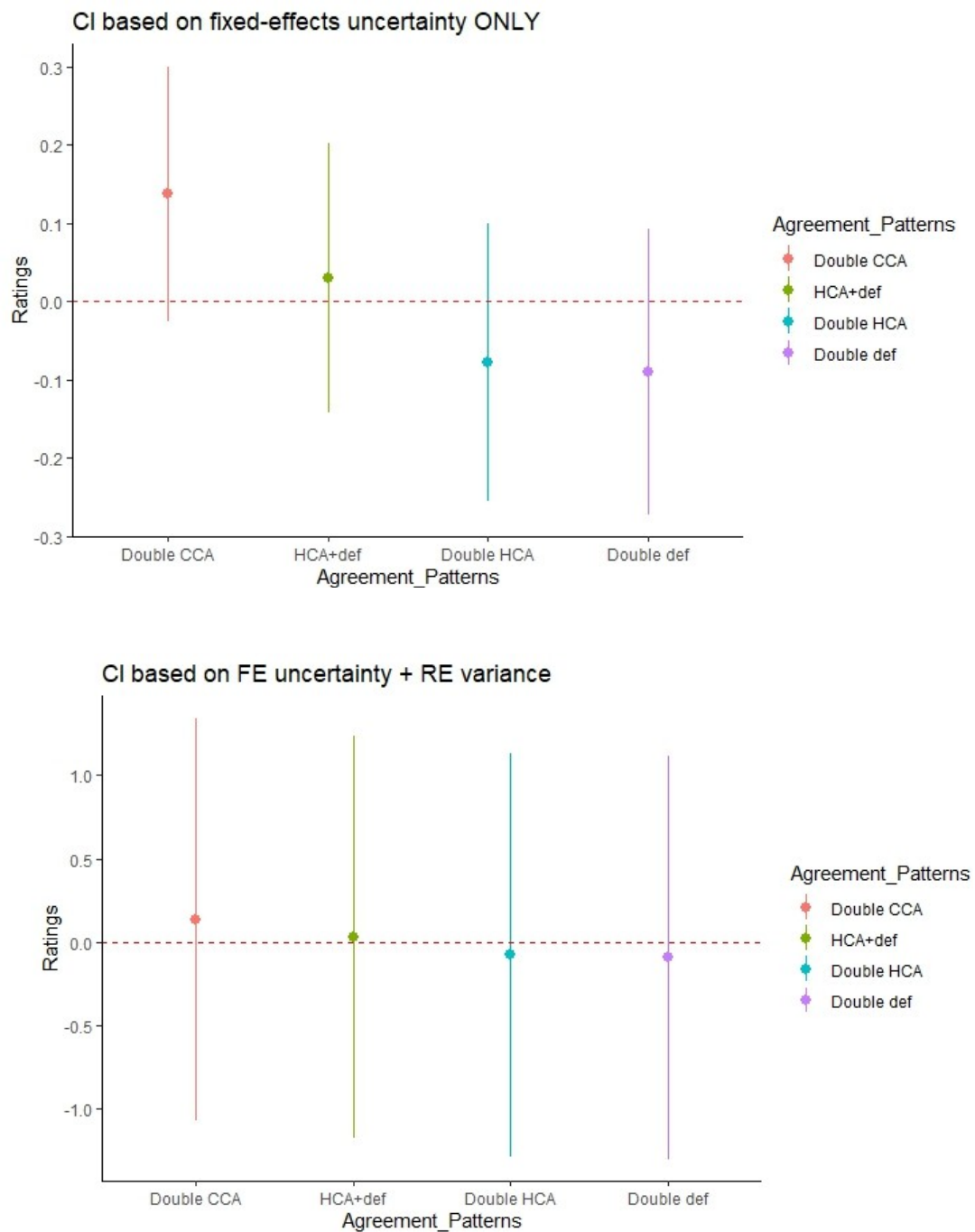


Figure 10: The coefficients' plot. Point estimates of the regression coefficients (dots) and their 95% CI.

These results call for a discussion on the interpretation of the significance of regression coefficients. The plot is created with 95% confidence intervals around the regression coefficients (red and blue dots depicting negative and positive ratings). The model fitted on the experimental data confirms positive ratings, hence acceptability, of the double CCA and HCA+def agreement patterns. Additionally, it provides confirmation of the double HCA as acceptable strategy, being under but not significantly different than the intercept (though on the edge), unlike the double default agreement pattern estimated as statistically significant from the reference level (total mean), and therefore unacceptable.

Next, we performed likelihood ratio comparisons (known as deviance tests) to perform significance tests with mixed models. For overall model performance, we test the overall effect of the predictor (the factor double agreement pattern), as well as random effects (participants). That includes model comparison for the significance of the predictor (fixed effects) and random effects. Results for the factor significance (model with the factor and null model) and random effects significance (model with random effects and model with no random effects) are presented in Appendix 7, confirming the significance of the model with the predictor, as well as the random effects significance. For the model comparisons regarding the significance of the fixed effects, a likelihood ratio test of the model with the agreement patterns effect against the model without the agreement patterns effect revealed a significant difference between models ($\chi^2(1) = 79.26, p < 0.0001$). There was also a significant effect of random effects, participants ($\chi^2(1) = 33.37, p < 0.0001$). This suggests that dropping the item random effects leads to a significant decrease in likelihood, hence the random effect is significant. With the last significance test performed, we got the pairwise differences of agreement patterns (Appendix 7, Table 4).

After communicating the regression coefficients, the overall model performance, and significance tests for the multilevel predictor, we are left with the last task of communicating the uncertainty of predictions with interval estimates trying to capture the population parameter. Figure 11b depicts where the actual values are likely to be with 95% confidence, with the summary tables in Appendix 7 (Tables 6 and 7). Together with previous results, plotted results provide statistical validation for the confirmations of the predictions and confirm trends observed in summary statistics.



Figures 11a-b: Computed confidence intervals on predictions.

6.2 *Discussing judgments response ratings as a function of agreement patterns*

We measured gradience in acceptability judgments in comprehension using acceptability rating task, an offline measure in visual modality. The aim of the study was to gather information about the internal grammar of BCMS speakers. In the experiment, the acceptability was measured of sentences with simple and sandwiched (double) agreement patterns under coordination. We wanted to know if there is a significant difference between the average acceptance rates of agreement patterns in the experimental conditions.

The acceptability of three double agreement pattern conditions and the unacceptability of the fourth condition in the experiment confirmed our four predictions. Namely, double CCA, double HCA, and HCA+def are confirmed as acceptable by BCMS speakers, and they follow the predictions and conclusions of Marušič & Nevins (2020) for Slovenian data. Next, following the syntactic accounts of Bošković (2009), which predicts preverbal HCA as unacceptable, and Murphy & Puškar (2018), who propose deactivation in the derivation, the double single conjunct agreement patterns (double HCA and double def) in this study are expected to be ungrammatical, considering strict syntactic approaches. While this is plausible regarding the results of the double default agreement pattern, estimated as unacceptable also by the distributed model, we don't see it with the other sandwiched single conjunct agreement pattern, double HCA. The deactivation property proposed by Murphy & Puškar (2018) could possibly account for double CCA and eventually HCA+def, but not double HCA. Hence, the two strict syntactic accounts when applied to double agreement strategies in BCMS in this study seem to be problematic, as the double agreement strategies demonstrate that independent probes target not only different agreement controllers but also one and the same agreement controller (HCA) on both sides' participles.

The double def agreement pattern would not be predicted as grammatical neither according to Marušič & Nevins (2020), as PostV def, on the Part 1 probe, is found unacceptable in the first place (for details, the reader is referred to Marušič & Nevins 2020). The PostV def pattern in BCMS is referred to as controversial. Murphy & Puškar (2018) judge the pattern as acceptable. On the other hand, Willer-Gold et al. (2016) reported it as unacceptable in an experimental study with BCMS speakers. The example of PostV def in Slovenian (Marušič et al. 2015: 23) was mentioned as acceptable, but in Marušič and Nevins (2020) as unacceptable empirically, and subsequently experimentally confirmed as unacceptable by Slovenian native speakers in the same study. In our study, the pattern was rated as degraded by the author of this article. In the experiment, the judgment of the article's author was confirmed. Namely, PostV def received lower rates amongst other simple, pre- and post-verbal, conjunction structures and was found to be statistically different than PreV def ($p < 0.00001$), see Appendix 5. The average rating of PreV def is 5.62, while PostV def has the average rating of 4.53. Therefore, when it comes to simple (pre- and post-verbal) &P configurations, PostV def cannot be taken as acceptable. Such results speak in support of comprehension and production studies of Willer-Gold et al. (2016; 2018) for BCMS and Marušič & Nevins (2020) for Slovenian, and contra Murphy & Puškar (2018). Important to mention, we do leave room for additional experimental confirmation and comparisons with pre- and post-verbal default conjunct agreement in other than participial sandwiched configurations. PostV def could not be comparable to the double (sandwiched) agreement patterns, as these are structurally more complexed, hence bear greater processing cost. This is apparent when we look at the average acceptance ratings for the PreV

def pattern (5.62), Sandwiched Simple (5.14), and the highest rated amongst coordinate sandwiched agreement patterns, double CCA (4.67), with double CCA being much below the first two.

Turning to sandwiched agreement strategies, the double default pattern reflects strategies of PostV def on Part 1 and PreV def on Part 2, while the HCA+def pattern reflects PostV HCA(CCA) on Part 1 (confirmed as the only one grammatical post-verbally in previous studies and here empirically), and the PreV def pattern on Part 2 (attested as robust cross-linguistically). Pairwise comparison between PreV def and PostV def reports significant difference, while pairwise comparison between HCA+def and double def was not significant. Having said that, the earlier prediction from Section 4 that the double def agreement pattern, with PostV def on Part 1, is to be judged poorly is born out. This might be expected given the three perspectives:

(i) PostV def was judged as acceptable by the syntactic account but unacceptable by the post-syntactic account and degraded, given my judgment. The double agreement pattern with PostV def on Part 1, double def, was confirmed as statistically unacceptable in the regression analysis;

(ii) For all double agreement patterns investigated in the study, Part 1 (with the post-verbal nominal target) agrees only with the linearly (hierarchically) closest controller NP₁, while with Part 2 (pre-verbal) as acceptable were found three agreement strategies, namely the linearly closest, hierarchically closest, and default agreement patterns (in line with the view that double agreement patterns reflect acceptable pre and post-verbal conjunction patterns);

(iii) The double def agreement pattern was the pattern with the largest number of extreme ratings (1 and 7) and fewer middle ratings than the other double agreement patterns. Whether the reason for that involves different preferences among speakers of four language varieties is out of the scope of the current study, and we leave it for future research. Possibilities we suggest being explored are to test the agreement patterns in a different context and additionally use speaker regional variants as a factor.

Another aspect of informativeness of our findings is that the sandwiched simple pattern, with ceiling acceptance rates, confirmed that such configurations are acceptable canonical configurations in BCMS grammar based on the introspection of the native speakers. The highest acceptability rating effect in coordinate sandwiched configurations was showed to have the double CCA pattern. Double CCA is direct evidence for the distributed agreement account, where looking at the surface structure on the first participle we see agreement with the conjunct before linearisation, and on the second participle we see agreement with the conjunct after linearisation. We are left with the double HCA agreement pattern, showed to be acceptable amongst BCMS speaker, a direct evidence contra the account of Murphy & Puškar (2018). Crucially, the confirmation of the phenomenon of first conjunct agreement pre-verbally in BCMS is strengthened by showing double HCA as being part of the BCMS grammar. However, we must notice that the pattern is close to be statistically different than the mean, more than the other two acceptable patterns. We interpret such results as a confirmation of the lower distribution among the BCMS speakers (previously confirmed for simple HCA under coordination in Arsenijević & Mitić 2019a for BCMS, and in Marušič et al. 2015 for Slovenian). We consider the fact that results of production experiments and rating experiments involve different domains, hence potential difference between production and perception data (see, for example, Willer-Gold 2023).

The premise that one of the conjuncts value gender of the verb, and number gets valued by the ConjP already speaks in support of split phi-feature probes. Experimental confirmation of individual conjunct agreement under sandwiched coordination regarding gender, discussed in this study, strengthens such postulation. In terms of two derivational choices, whether default agreement is chosen or not, given the proviso that it can only be chosen when the subject &P on the surface c-commands the participle, is the question that we leave for future research. It would complement the adequacy of the distributed model – why the default agreement, following the No-peeking grammar, is found pre-verbally but not post-verbally. In other words, why is the No-peeking grammar constrained? As previously mentioned in Nevins & Weissner (2019: p. 12), syntactic accounts such as Bošković (2009) and van Koppen (2005), without further elaboration, explain for the agreement patterns, where pre-verbal orders trigger resolved agreement, but post-verbal orders trigger HCA. The specifier competes with the conjunction phrase as an agreement target only when they are both c-commanded by the agreeing head. When the agreeing head is c-commanded by the conjunction phrase, no competition arises. The claim was also adopted in Aljović & Begović (2016), who further postulate the link with the case assignment domain and HCA as the more specified goal to account for the cases where the HCA but not the default masculine was found acceptable.

Such a line of reasoning leads us to establish a firmer claim: the default specification arises when the conflicting ϕ features' overload cause a speaker to resort to the least complex option in terms of processing demands. If so, default agreement bears the least processing cost, hence reflects the shortest time to respond, in experimental settings. This was one of the questions Palmović & Willer-Gold (2016) asked in their study. Different effects could play a role in deriving an agreement pattern: distance effect, syntactic complexity, syntactic ambiguity, extrinsic memory load. Since language comprehension processes occur simultaneously, numerous activations may burden capacities (in terms of Just & Carpenter 1992). If the working memory capacity involved, HCA bears the greatest processing cost. The findings of Palmović & Willer-Gold (2016) are leaned towards no involvement of higher cognitive mechanisms and more of a linguistic domain involved. We commit ourselves to explore HCA in greater depth, along with the other two strategies, in a real-time sentence comprehension set up such as a self-paced reading experiment, given a different syntactic configuration under coordination (Adj-AUX-&P-V, Adj-NP1-AUX-&NP2-V, Adj-NP1-AUX-Part1-&NP2-Part2), considering Left-Branch-Extraction (LBE) and Coordinate Structure Constraint Violation (CSCV; for similar recent empirical studies see, e.g., Gračanin-Yuksek & Arsenijević 2017 and Shen 2023). We also aim to check the three double agreement patterns from the perspective of intra-speaker variability, given simultaneous alternatives in a forced choice study with different types of coordination.

7 Conclusion

Using introspective judgments, the study aimed to estimate the acceptability of agreement strategies under sandwiched coordination with BCMS data. An acceptability rating experiment was conducted, involving a visual sentence comprehension on the Likert scale, set up to investigate inter-variability of agreement strategies that can be accounted for as sensitivity to the linear proximity, namely directionality happening post-syntactically. We reported on the results of the experiment, looking at the verbal inflection on participles under sandwiched conjunction in pluperfect tense in BCMS, where

number and gender markings on Parts controlled by the coordinate nouns are in passive structures. The hypothesis that participles found on each side of coordination phrase act independently and agree with different/same conjuncts was confirmed. Specifically, the sandwiched configurations in this study mirror exactly those agreement patterns found pre- and post-verbally. The particular sandwiched construction showed to be part of the BCMS grammar.

The second issue was of feature bundling and separate vs. simultaneous agreement of number and gender. In light of Pesetsky & Torrego (2007) assumptions of the probe responsible for bearing agreement suffixes, we have adopted the central characteristic of a probe is not (un)interpretability but rather the ability to undergo valuation. Valuation is not to be understood in terms of supplying a value for a given attribute (in Bejar's 2003 terms, a category), but rather in terms of transferring features from a goal to a probe (copying). Unlike the Bošković's (2009) account, in the Marušič et al.'s (2015) account, the probe is argued to operate independently with respect to number and gender. In Bošković's "Secondary Agree", valuing gender on the target assumes probing twice inside &P. Marušič et al.'s "Split-Probe" or "No-Default", probing occurs in having different goals for number and gender. Under both mechanisms, the probe searches for matching number features, and finds its number feature on &P and its gender feature on NP₁ or NP₂.

We showed the drawbacks of the Bošković's account, which include FCA pre-verbally and subsequently sandwiched as impossible. On the other side, Marušič et al. (2007), Marušič & Nevins (2010), and consequently Marušič et al. (2015) and Marušič & Nevins (2020), using split-probe mechanism comprising syntactic and post-syntactic stages, arrive to the availability of the three separate agreement strategies of (double) closest conjunct agreement, highest conjunct agreement, and agreement with ConjP. They experimentally showed that all three grammars are available to speakers and used according to preferences, therefore their distribution is different.

The approach concerning varying orders of the operations Agree and Merge inside the conjunct phrase, used in Murphy & Puškar (2018), is challenged by sandwiched agreement strategies, since deactivation wouldn't make the patterns possible under the assumption that probes act independently. The present results of this study cannot be straightforwardly accommodated under a strict syntactic approach and are more in line with the post-syntactic approach, favouring the suggestions of the Marušič & Nevins (2020) account for sandwiched configurations in Slovenian to be extended to BCMS.

Abbreviations

1/2/3 = first/second/third person
 AUX = auxiliary
 CCA = closest conjunct agreement
 def = default agreement
 DCA = distant conjunct agreement
 F = feminine gender
 FCA = first conjunct agreement
 Gen = gender

HCA = highest conjunct agreement
 LCA = last conjunct agreement
 M = masculine gender
 N = neuter gender
 Num = number
 Part = participle
 PL = plural number
 SG = singular number

Supplementary file

Supplementary materials include complete stimuli list (control and critical items in Appendix 1, filler items in Appendix 2) and supplementary analyses scripts (Appendices 3-7).

Acknowledgments

Invaluable gratitude goes to Lanko Marušič and Rok Žaucer from the Center for Cognitive Science of Language, University of Nova Gorica. The topic sparked my interest during research work on coordination with Lanko and continued during the paper writing seminar with Rok. I am also grateful to Lanko for comments and suggestions on numerous earlier drafts of this paper. The research at its early stage (the pilot study) was presented at two conferences in 2022, namely University of Alabama Linguistics Conference and Bogaziçi University Student Conference on Linguistics, where I received many excellent questions and valuable suggestions. The work was partly presented at Slavic Linguistics Colloquium in June 2022, and I thank the audience for feedback and comments. Three anonymous reviewers put considerable effort in constructive comments and recommendations, which I am grateful for. Their feedback greatly improved the article. Most of all, I thank experiment participants for their valuable judgments during several stages of the study. All mistakes are my responsibility alone.

Competing interest

The author has no competing interests to declare.

References

- Aljović, Nadira & Begović, Muamera. 2016. Morphosyntactic aspects of adjectival and verbal first-conjunct agreement. *Journal of Slavic Linguistics* 24(1). 7-39. DOI: [10.1353/jsl.2016.0004](https://doi.org/10.1353/jsl.2016.0004)
- Anwyl-Irvine, Alexander L. & Massonnié, Jessica & Flitton, Adam & Kirkham, Natasha & Evershed, Jo K.. 2018. Gorilla experiment builder. <https://gorilla.sc/>
- Arregi, Karlos & Nevins, Andrew. 2012. *Morphotactics: Basque auxiliaries and the structure of spellout*. Vol. 86 (Studies in Natural Language and Linguistic Theory). Dordrecht: Springer. DOI: <https://doi.org/10.1007/978-94-007-3889-8>
- Arsenijević, Boban & Mitić, Ivana. 2016a. On the number-gender (in)dependence in agreement with coordinated subjects. *Journal of Slavic linguistics* 24(1). 41-69. DOI: <https://ojs.ung.si/index.php/JSL/article/view/363>
- Arsenijević, Boban & Mitić, Ivana. 2016b. Effect of animacy and agentivity on the processing of agreement in Serbo-Croatian. In Halupka-Rešetar Sabina & Martínez-Ferreiro, Silvia (eds.), *Studies in language and mind: Selected papers from third Novi Sad Workshop on psycholinguistics, neurolinguistic and clinical linguistic research*, 41-77. Novi Sad: University of Novi Sad, Faculty of Philosophy. <https://digitalna.ff.uns.ac.rs/sites/default/files/db/books/978-86-6065-359-0.pdf>
- Arsenijević, Boban & Willer-Gold, Jana & Aljović, Nadira & Čordalija, Nermina & Kresić-Vukosav, Marijana & Leko, Nedžad & Malenica, Frane & Marušič, Franc Lanko & Milićev, Tanja & Milićević, Nataša & Mišmaš, Petra & Mitić, Ivana & Peti-Stantić, Anita & Stanković, Branimir & Tušek, Jelena

- & Nevins, Andrew. 2019. Elided clausal conjunction is not the only source of closest-conjunct agreement: A picture-matching study. *Syntax* 23(1). 78-104. DOI: <https://doi.org/10.1111/synt.12171>
- Aoun, Joseph & Benmamoun, Elabbas & Sportiche, Dominique. 1994. Agreement, word order, and conjunction in some varieties of Arabic. *Linguistic Inquiry* 25. 195-220.
- Bates, Douglas & Martin Mächler & Ben Bolker & Steve Walker. 2015. Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67.1. DOI: <https://doi.org/10.18637/jss.v067.i01>
- Begović, Muamera & Aljović, Nadira. 2015. Accounting for agreement patterns in coordinate noun phrases with a shared modifier. In Ziková, Markéta & Caha, Pavel & Dočekal, Mojmír (eds.), *Slavic languages in the perspective of formal grammar: Proceedings of FDSL 10.5*, 39-59. Frankfurt: Peter Lang.
https://api.pageplace.de/preview/DT0400.9783653053487_A31444473/preview-9783653053487_A31444473.pdf
- Béjar, Susana. 2003. *Phi-syntax: A theory of agreement*. Toronto, ON: University of Toronto dissertation.
- Benmamoun, Elabbas & Bhatia, Archana & Polinsky, Maria. 2009. Closest Conjunct Agreement in Head Final Languages. *Linguistic Variation Yearbook* 9(1). 67-88. DOI: [10.1075/livy.9.02ben](https://doi.org/10.1075/livy.9.02ben)
- Bhatt, Rajesh & Walkow, Martin. 2013. Locating agreement in grammar: An argument from agreement in conjunctions. *Natural language and linguistic theory* 31(4). 951-1013. DOI: <https://doi.org/10.1007/s11049-013-9203-y>
- Bojović, Draga. 2003. Neki slučajevi kongruencije koordinirane subjekatske sintagme s preponiranim i postponiranim predikatom u srpskom jeziku [Cases of Agreement of Coordinated Subject Phrases with Preposed and Postposed Predicate in the Serbian language]. *Srpski jezik* 8(1-2). 539-555.
- Bošković, Željko. 2009. Unifying first and last conjunct agreement. *Natural Language & Linguistic Theory* 27(3). 455-96. DOI: <https://doi.org/10.1007/s11049-009-9072-6>
- Bošković, Željko. 2010. Conjunct-sensitive agreement: Serbo-Croatian vs Russian. In Zybatow, Gerhild & Dudchuk, Philip & Minor, Serge & Pshehotskaya, Ekaterina (eds.), *Formal Studies in Slavic Linguistics: Proceedings of FDSL 7.5*, 31-48. Peter Lang: Frankfurt am Main.
- Citko, Barbara. 2004. Agreement asymmetries in coordinate structures. In Arnaudova, Olga & Browne, Wayles & Rivero, Maria Luisa & Stojanović, Danijela (eds.), *Formal Approaches to Slavic linguistics: The Ottawa Meeting*, 91-108. Ann Arbor: Michigan Slavic Publications.
- Citko, Barbara. 2018. Complementizer agreement with coordinated subjects in Polish. *Glossa: a journal of general linguistics* 3(1). 124. 1-25. DOI: <https://doi.org/10.5334/gjgl.588>
- Chomsky, Noam. 1995. *The Minimalist Program*. (Current Studies in Linguistics 28). Cambridge, MA: MIT Press.

- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Martin, Roger & Michaels, David & Uriagereka, Juan & Keyser, Samuel Jay (eds.), *Step by step: Essays on Minimalist syntax in honour of Howard Lasnik*. 89-155. MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In Michael Kenstowicz (ed.), *Ken Hale: A Life in Language*. 1-52. Cambridge, MA: MIT Press. DOI: [10.7551/mitpress/4056.003.0004](https://doi.org/10.7551/mitpress/4056.003.0004)
- Corbett, Greville G. 1983a. *Hierarchies, targets, and controllers: Agreement patterns in Slavic*. London: Croom Helm.
- Corbett, Greville G. 1983b. Resolution rules: Agreement in person, number, and gender. In Gazdar, Gerald & Klein, Ewan & Pullum, Geoffrey (eds.), *Order, concord, and constituency*, 175-206. Dordrecht: Foris.
- Crone, Phil. 2016. Finnish first conjunct agreement & the direction of Agree. *Finno-Ugric Language and Linguistics* 5(2). 2-45.
- Čordalija, Nermina & Bešić, Amra & Jovović, Ivana & Marijanović, Nevenka & Perković, Lidija & Šaljić, Midhat & Telalagić, Dženana & Leko, Nedžad. 2016. Grammars of participle agreement with conjoined subjects in Bosnian/Croatian/Serbian. *Journal of Slavic linguistics* 24(1). 71-112. DOI: <https://ojs.ung.si/index.php/JSL/article/view/365>
- Čordalija, Nermina & Jovović, Ivana & Leko, Nedžad. 2020. Postverbal conjoined subjects and closest conjunct agreement in Bosnian/Croatian/Serbian: An experimental study. *Suvremena lingvistika* 46(89). 25-47. DOI: <https://doi.org/10.22210/suvlin.2020.089.02>
- Despić, Miloje. 2017. Investigations in mixed agreement: Polite plurals, hybrid nouns, and coordinate structures. *Morphology* 27(3). 253-310. DOI: <https://doi.org/10.1007/s11525-017-9301-3>
- Drummond, Alex. 2011. IbexFarm. Version 0.2.7. spellout.net/ibexfarm
- Embick, David & Noyer, Rolf. 2001. Movement operations after syntax. *Linguistic Inquiry* 32(4). 555-595. DOI: <https://doi.org/10.1162/002438901753373005>
- Embick, David & Noyer, Rolf. 2012. Distributed morphology and the syntax/morphology interface. In Ramchand, Gillian & Reiss, Charles (eds.), *The Oxford handbook of linguistic interfaces*, 289-324. Oxford Academic. DOI: <https://doi.org/10.1093/oxfordhb/9780199247455.013.0010>
- Franks, Steven & Willer-Gold, Jana. 2015. Agreement strategies with conjoined subjects in Croatian. In Witkos, Jacek & Jaworski, Sylwester (eds.), *New insights into Slavic linguistics*, 91-115. Frankfurt am Main: Peter Lang. DOI: <https://doi.org/10.3726/978-3-653-04359-4>
- Gračanin-Yuksek, Martina & Arsenijević, Boban. 2017. Auxiliary clitics in coordinated subjects: AGREE–SPLIT–REPEAT. Slides from an invited talk at *Formal Descriptions of Slavic Linguistics 12.5*, University of Nova Gorica, Nova Gorica, Slovenia.
- Hiraiwa, Ken. 2005. *Dimensions of symmetry in syntax: agreement and clausal architecture*. Cambridge, MA: MIT dissertation.

Halle, Morris & Marantz, Alec. 1993. Distributed Morphology and the pieces of inflection. In Hale, Kenneth & Keyser, Samuel Jay (eds.), *The view from building 20: Linguistic Essays in Honor of Sylvian Bromberger*, 111-176. Cambridge, MA: MIT Press.

Halle, Morris & Marantz, Alec. 1994. Some key features of Distributed Morphology. *MIT Working papers in linguistics 21: Papers on Phonology and Morphology*. 275-288.

Halle, Morris. 1997. Distributed Morphology: Impoverishment and Fission. *MIT Working Papers in Linguistics 30: PF Papers at the Interface*. 125-149.

Just, Marcel Adam & Carpenter, Patricia A. 1992. A capacity theory of comprehension: Individual differences in working memory. *Psychological Review* 99(1). 122-149.

DOI: <https://doi.org/10.1037/0033-295x.99.1.122>

Kobayashi, Akiko. 2022. Past participle agreement in French and Italian: A two-Agree analysis. *Glossa: a journal of general linguistics* 7(1). 1-41. DOI: <https://doi.org/10.16995/glossa.5830>

Leko, Nedžad. 2015a. Slaganje predikata sa koordiniranim imeničkim sintagmama u bosanskom jeziku [Agreement of predicates with coordinated noun phrases in the Bosnian language]. *Radovi Filozofskog fakulteta u Sarajevu* 18. 164-179.

Leko, Nedžad. 2015b. Analiza grešaka u jezičkoj produkciji ispitanika u eksperimentalnoj studiji slaganja predikata sa koordiniranim subjektom u bosanskom/hrvatskom/srpskom jeziku [An analysis of performance errors in an experimental study of participle agreement with conjoined subjects in Bosnian/Croatian/ Serbian]. *Pismo: časopis za jezik i književnost* 13. 43-55.

Marušič, Franc & Nevins, Andrew & Saksida, Amanda. 2007. Last-conjunct agreement in Slovenian. In Compton, Richard & Goledzinowska, Magdalena & Savchenko, Ulyana (eds.), *Formal Approaches to Slavic Linguistics 15: The Toronto Meeting*, 210-227. Ann Arbor: Michigan Slavic Publications.

Marušič, Franc & Nevins, Andrew. 2010. Two types of neuter: Closest-conjunct agreement in the presence of '5&Ups'. In Browne, Wayles & Cooper, Adam & Fisher, Alison & Kesici, Esra & Predolac, Nikola & Zec, Draga (eds.), *Formal approaches to Slavic linguistics 18: The second Cornell meeting*, 301-317. Ann Arbor, MI: Michigan Slavic Publications.

Marušič, Franc & Nevins, Andrew Ira & Badecker, William. 2015. The grammars of conjunction agreement in Slovenian. *Syntax* 18(1). 39-77. DOI: <https://doi.org/10.1111/synt.12025>

Marušič, Franc & Nevins, Andrew. 2020. Distributed agreement in participial sandwiched configurations. In Smith, Peter W. & Mursell, Johannes & Hartmann, Katharina (eds.), *Agree to Agree: Agreement in the Minimalist programme*, 179-198. Berlin: Language Science Press.

Marušič, Franc Lanko & Shen, Zheng. 2021. Gender agreement with exclusive disjunction in Slovenian. *Acta Linguistica Academica* 68(4). 516-535.

DOI: <https://doi.org/10.1556/2062.2021.00443>

- Mitić, Ivana & Arsenijević, Boban. 2019a. Structural ambiguity and optionality of agreement patterns in Bosnian/Croatian/Serbian conjunct agreement. *Glossa: a journal of general linguistics* 4(1): 6. DOI: <https://doi.org/10.5334/gjgl.582>
- Mitić, Ivana & Arsenijević, Boban. 2019b. Plural conjuncts and syncretism facilitate gender agreement in Serbo-Croatian: Experimental evidence. *Frontiers in Psychology* 10 (942). DOI: <https://doi.org/10.3389/fpsyg.2019.00942>
- Moskovljević, Jasmina. 1983. O kongruenciji glagola sa naporednom imeničkom sintagmom [On the congruence of verbs with consecutive noun phrases]. *Psihologija: časopis Saveza društava psihologa Srbije* 16(4). 72-78.
- Munn, Alan. 1999. First conjunct agreement: Against a clausal analysis. *Linguistic Inquiry* 30(4). 643-668. DOI: <https://doi.org/10.1162/002438999554246>
- Murphy, Andrew & Puškar, Zorica & Guzmán Naranjo, Matías. 2018. Gender encoding on hybrid nouns in Bosnian/Croatian/Serbian: Experimental evidence from ellipsis. In Lenertová, Denisa & Meyer, Roland & Šimík, Radek & Szucsich, Luka (eds.), *Advances in formal Slavic linguistics* 2016, 313-336. Berlin: Language Science Press. DOI: [10.5281/zenodo.2546440](https://doi.org/10.5281/zenodo.2546440)
- Murphy, Andrew & Puškar, Zorica. 2018. Closest conjunct agreement is an illusion. *Natural Language & Linguistic Theory* 36(1). 1207-1261. DOI: <https://doi.org/10.1007/s11049-017-9396-6>
- Nevins, Andrew. 2011. Multiple agree with clitics: Person complementarity vs. omnivorous number. *Natural Language & Linguistic Theory* 29(4). 939-971. DOI: <https://doi.org/10.1007/s11049-011-9150-4>
- Nevins, Andrew. 2018. Copying and resolution in South Slavic and South Bantu conjunct agreement. In Petrosino, Roberto & Cerrone, Pietro & van der Hulst, Harry (eds.), *From sounds to structures: Beyond the veil of Maya*, 391-408. Berlin, Boston: De Gruyter Mouton. DOI: <https://doi.org/10.1515/9781501506734-013>
- Nevins, Andrew & Weisser, Philipp. 2019. Closest conjunct agreement. *Annual Review of Linguistics* 5. 219-241. DOI: <https://doi.org/10.1146/annurev-linguistics-011718-012708>
- Palmović, Marijan & Willer-Gold, Jana. 2016. Croatian mixed-gender conjunct agreement: An ERP study. *Journal of Slavic linguistics* 24(1). 137-160. DOI: <https://ojs.ung.si/index.php/JSL/article/view/367>
- Pesetsky, David & Torrego, Esther. 2007. The syntax of valuation and the interpretability of features. In Karimi, Simin & Samiian, Vida & Wilkins, Wendy K. (eds.), *Phrasal and clausal architecture: Syntactic derivation and interpretation*, 262-294. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/la.101.14pes>
- Prażmowska, Anna. 2016. Full agreement with coordinate subjects in Polish: Gender resolution rules revisited. *Roczniki Humanistyczne* 64(11). 71-86. DOI: <https://doi.org/10.18290/rh.2016.64.11-5>

- Puškar, Zorica & Murphy, Andrew. 2015. Closest-conjunct agreement in Serbo-Croatian: A rule-ordering account. In Assmann, Anke & Bank, Sebastian & Georgi, Doreen & Klein, Timo & Weisser, Philipp & Zimmermann, Eva (eds.). *Linguistische Arbeits Berichte 92: Topics at InfL*, 441-482. Leipzig: Institut für Linguistik der Universität Leipzig.
https://www.philol.uni-leipzig.de/fileadmin/Fakult%C3%A4t_Philol/Linguistik/Forschung/LAB/LAB_92/LAB92_15_puskar_murphy.pdf
- Puškar, Zorica. 2017. *Hybrid agreement: Modelling variation, hierarchy effects, and ϕ -feature mismatches*. Leipzig, Germany: Universität Leipzig dissertation.
- Puškar, Zorica. 2018. Interactions of gender and number agreement: Evidence from Bosnian/Croatian/Serbian. *Syntax* 21(3). 275-318. DOI: <https://doi.org/10.1111/synt.12154>
- R Core Team. 2020. R: A language and environment for statistical computing. R Foundation for statistical computing. Vienna, Austria. <https://www.R-project.org/>
- RStudio Team. 2019. RStudio: Integrated development for R. RStudio, Inc. Boston, MA. <http://www.rstudio.com/>
- Shen, Zhen. 2023. Conjunction agreement and the coordinate structure constraint. *Glossa: a journal of general linguistics* 8(1). 1-32. DOI: <https://doi.org/10.16995/glossa.6382>
- van Koppen, Johanna Maria. 2005. *One probe, two goals: Aspects of agreement in Dutch dialects*. Leiden, Netherlands: Universiteit Leiden dissertation.
- van Koppen, Marjo. 2007. Agreement with coordinated subjects: A comparative perspective. *Linguistic Variation Yearbook* 7(1). 121-161. DOI: <https://doi.org/10.1075/livy.7.05kop>
- Wechsler, Stephen & Zlatić, Larisa. 2003. *The many faces of agreement*. (Stanford Monographs in Linguistics, none.) Stanford, CA: Center for the Study of Language and Information (CSLI) Publications.
- Westfall, Jacob & Kenny, David A. & Judd, Charles M. 2014. Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General* 143(5). 2020-2045. <https://jakewestfall.shinyapps.io/crossedpower/>
- Willer-Gold, Jana & Arsenijević, Boban & Batinić, Mia & Čordalića, Nermina & Kresić, Marijana & Leko, Nedžad & Marušić, Franc Lanko & Milićev, Tanja & Milićević, Nataša & Mitić, Ivana & Nevins, Andrew & Peti-Stantić, Anita & Stanković, Branimir & Šuligoj, Tina & Tušek, Jelena. 2016. Conjunct agreement and gender in South Slavic: From theory to experiments to theory. *Journal of Slavic Linguistics* 24(1). 187-224. DOI: <https://ojs.ung.si/index.php/JSL/article/view/370>
- Willer-Gold, Jana & Arsenijević, Boban & Batinić, Mia & Becker, Michael & Čordalića, Nermina & Kresić, Marijana & Leko, Nedžad & Marušić, Franc Lanko & Milićev, Tanja & Milićević, Nataša & Mitić, Ivana & Peti-Stantić, Anita & Stanković, Branimir & Šuligoj, Tina & Tušek, Jelena & Nevins, Andrew. 2018. When linearity prevails over hierarchy in syntax. *The Proceedings of the National Academy of Science (PNAS)* 115(3). 495-500. DOI: [10.1073/pnas.1712729115](https://doi.org/10.1073/pnas.1712729115)

Willer-Gold, Jana. 2023. Locus and timing of gender resolution: Probes, goals and predictions. *Glossa: a journal of general linguistics* 8(1). 1-45. DOI: <https://doi.org/10.16995/glossa.8939>

Willim, Ewa. 2012. Concord in Polish coordinate NPs as agree. In Ziková, Markéta & Dočekal, Mojmír (eds.), *Slavic languages in formal grammar: Proceedings of FDSL 8.5, Brno 2010*, 233-253. Frankfurt am Main: Peter Lang.